Semantic-Guided Mesh-to-Mesh Deformation

Study Project Report

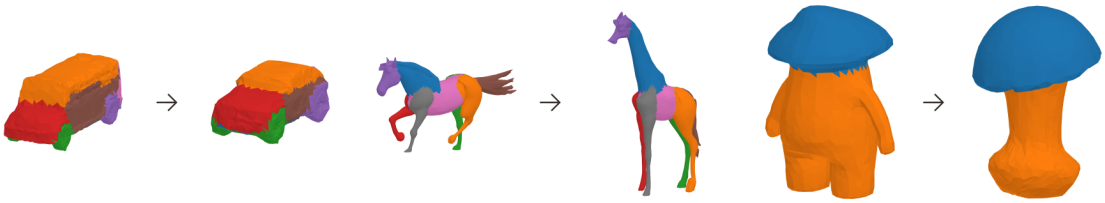
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Abstract

This project studies mesh-to-mesh deformation: given a source 3D shape, deform it so that it matches a target shape while preserving the identity of meaningful parts. A purely global surface loss, such as Chamfer distance, can make two shapes overlap geometrically while still mixing parts: a leg can move toward a trunk, or a head can collapse into the body, if that reduces the average point distance. This project's central idea is to add local, part-aware supervision from PartField segmentations. Source and target meshes are segmented into unsupervised semantic buckets; deformation is then optimized with a global Chamfer term, a per-bucket local Chamfer term, and regularizers that discourage folding and excessive local distortion. The project began as an extension of MeshUp, a score-distillation mesh deformation method driven by a 4.3B-parameter diffusion model, before being redirected into direct mesh-to-mesh deformation in 3D; that redirection is documented and explained in Appendix B.

The pipeline is evaluated on DenseCorr3D and on a self-created collection of SAM3D-reconstructed meshes. The headline result is a graph-geodesic, same-pairs comparison against DenseMatcher's own predicted correspondences: across all 23 available categories, with zero PartField-level selection, this pipeline reaches mean AUC 0.824 against DenseMatcher's 0.672, winning 21 of 23. A stricter no-preparation subset gives the same positive signal: among 12 categories without assisted orientation/preparation, the same range-averaged protocol still beats DenseMatcher in 10 categories. This comparison, its methodology, and its scope are the central subject of Section 7. Surface-quality evaluation on self-created SAM3D mesh pairs is reported in Section 8. The report also documents concrete failure modes – orientation mistakes, unstable part buckets, topology mismatch, and metric sensitivity to thin evaluation regions – with a full root-cause trace for the hardest produce categories.

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1. Introduction

This report describes a project on 3D mesh deformation. In ordinary images, the data lives on a flat grid of pixels. In 3D, the object has geometry: points have coordinates (x, y, z), surfaces have normals and curvature, and distances can be measured either through space or along the surface itself. A useful deformation method should do more than move points until two shapes look similar. It should keep the structure of the object coherent.

The motivating example is animal deformation. Suppose the source is a horse mesh and the target is a giraffe mesh. A good result should stretch the neck, elongate the legs, and adjust the body without losing the idea that legs match legs, head matches head, and body matches body. If we only minimize global Chamfer distance, the optimizer sees an unordered set of surface samples. It can reduce the average distance between the two surfaces while still matching the wrong parts. This is the central ambiguity the project tries to reduce.

The pipeline developed here adds part-level structure by using PartField. PartField produces unsupervised segmentation buckets for a mesh: not class labels such as "leg" or "head", but consistent local regions that in most cases behave like semantic parts. These buckets are used as local correspondence constraints. The deformation is still supervised by the target geometry, but it is now asked to match source bucket k to target bucket k instead of treating all surface points as interchangeable.

This project did not start from the pipeline described here; an earlier, image-semantic approach built on MeshUp was tried first and deliberately redirected once its limits became clear. That earlier direction, with full numbers, is documented in Appendix B.

2. Motivation and Problem Setting

2.1 Why this problem matters

3D shape deformation is useful whenever a known object must be adapted to a new shape while remaining editable and interpretable. In animation, a character mesh may need to follow a new pose. In graphics, a template mesh may need to fit a scanned object. In reconstruction, a generated mesh may be close to the target object but still need geometric refinement. In all of these cases, the mesh is not just a surface to be made close to another surface. It is also a structured object whose parts should remain meaningful.

The motivating difficulty is that geometry and semantics are not the same thing. A global surface distance can say that two shapes are close even when the correspondence is wrong. For example, if a source animal has four legs and a target animal also has four legs, a global loss does not know which leg should match which leg. If a target has a long neck or trunk, a global loss may pull a nearby source region into that space even if the source region semantically belongs to the body. The shape may look acceptable from one view while the underlying correspondence is poor.

This is especially important for meshes. A mesh carries topology, adjacency, and vertex identity. If the deformation is later used for texture transfer, animation, interpolation, or correspondence analysis, it matters whether the right vertices went to the right target regions. A pure surface-fit objective cannot guarantee this.

2.2 Why PartField is a useful prior

The ideal supervision would be dense correspondence: for every point on the source surface, the method would know the matching point on the target surface. This is rarely available outside benchmarks. Manual part labels are easier than dense correspondences, but still expensive and not available for arbitrary generated or reconstructed meshes. PartField [7] sits between these two extremes. It gives a mesh decomposition into repeated local regions without manual labels. The buckets are not perfect semantic labels, but they often align with object parts well enough to guide optimization.

The important practical advantage is that PartField can be used on the same kinds of meshes that the deformation pipeline already consumes. This means the project can use a single principle across benchmark

animals, SAM3D animals, vehicles, tools, and produce categories: deform globally, but ask local regions to match local regions. The method is therefore not tied to one category-specific skeleton or a hand-written list of parts. Figure 1 shows what this decomposition looks like in practice on a compatible pair.

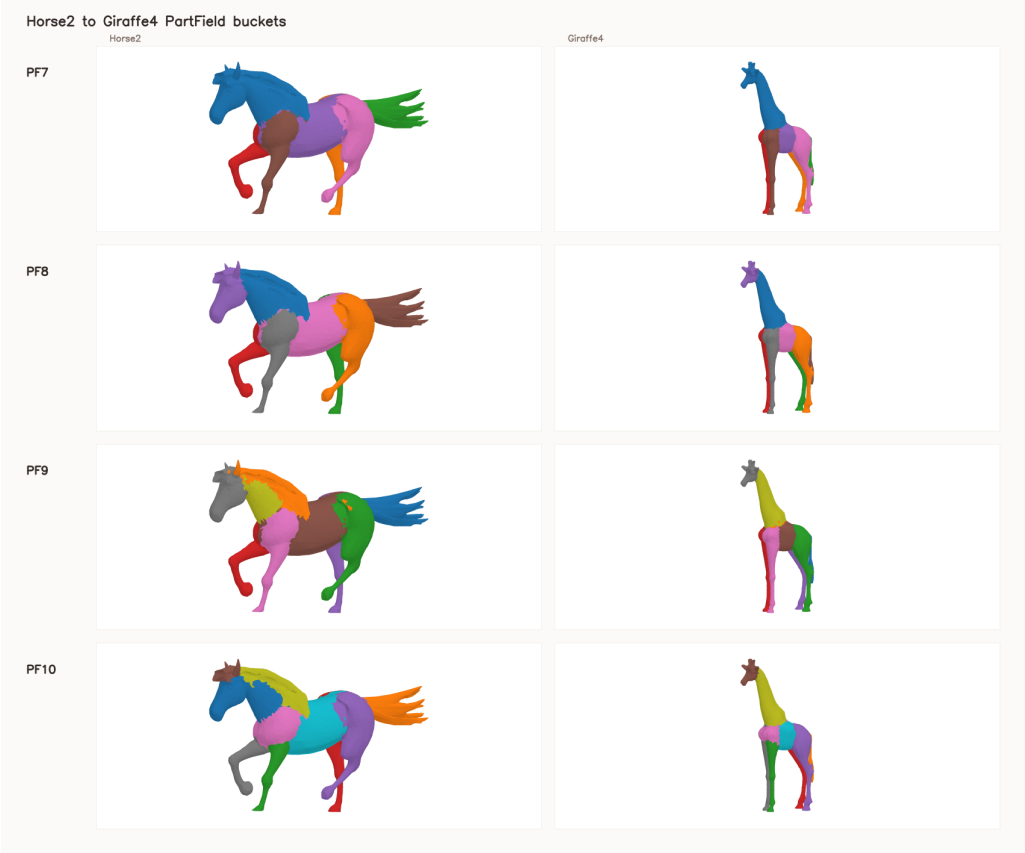


Figure 1. PartField decomposition for Horse2 → Giraffe4, PF7–PF10.

2.3 Research questions

The experiments are organized around four research questions:

1. Does local PartField Chamfer improve semantic alignment compared with judging the result only by global surface proximity, *and does it hold up against a real correspondence method scored on the exact same pairs*?
2. Which PartField granularity is most useful, and is any single granularity stable enough across categories to be trusted as a fixed hyperparameter, rather than needing to be re-tuned per category?
3. Do the deformed meshes remain geometrically clean when the local semantic loss is added, and does that hold across very different object domains?
4. What kinds of shapes cause the pipeline to fail, and are those failures explainable?

These questions also define the scope of the report. The project is a deformation pipeline evaluated with correspondence metrics. The wording is important because DenseMatcher predicts correspondence directly, whereas this project optimizes a source mesh until it becomes close to a target mesh. DenseCorr3D annotations are used as an independent evaluator of the final deformation, and Section 7 is explicit about what a comparison against a direct correspondence method can and cannot claim.

2.4 What counts as success

A successful deformation should satisfy three conditions. First, the output surface should be close to the target. This is measured by Chamfer, F@0.02, and robust percentile distances. Second, the output should preserve local semantic identity. On DenseCorr3D this is measured by geodesic error, AUC, and same-bucket agreement. Third, the result should remain a plausible mesh. This is addressed by regularization and inspected through qualitative figures and video showcases.

No single metric captures all three conditions. A result can have low Chamfer but poor semantic alignment; it can have high AUC but a few severe tail failures; or it can match parts well but look visually stretched. For that reason multiple metrics are taken into account.

2.5 Project goals

Given this setting, the project has three concrete goals:

1. Build a practical target-guided mesh deformation pipeline using global geometry, local part guidance, and regularization.
2. Evaluate whether the deformed meshes are geometrically close to their targets.
3. Evaluate whether the deformation preserves semantic correspondence, especially on DenseCorr3D where independent semantic-group annotations are available for evaluation, and directly against DenseMatcher's own predictions on the same pairs.

3. Basic Concepts

3.1 3D shapes, point clouds, and meshes

A 3D shape is a geometric object embedded in three-dimensional space. In computation, the same shape can be represented in several ways. A *point cloud* is a set of points sampled from the surface. It is simple and flexible, but it does not explicitly say which points are connected to which other points. A *mesh* adds this missing structure. In this project the mesh is a triangular mesh, written as

$$M = (V, F),$$

where $V \in \mathbb{R}^{n\times 3}$ is the set of vertex coordinates and F is a list of triangular faces, each face indexing three vertices. The faces define the surface. Because connectivity is known, a mesh supports surface normals, edge lengths, local neighborhoods, geodesic distances, and deformation regularizers.

This distinction matters. Chamfer distance can be computed on point samples, but deformation is applied to a mesh. If a method moves vertices without respecting mesh structure, the surface can fold, stretch, tear visually, or develop distorted triangles. The pipeline therefore optimizes surface matching and also regularizes the deformation field.

3.2 Chamfer distance

Chamfer distance is a standard way to compare two point sets. Let P be points sampled from the deformed source mesh and Q be points sampled from the target mesh. The symmetric L1 Chamfer distance used for evaluation in this project is

$$d_{\text{Chamfer}}(P, Q) = \frac{1}{|P|}\sum_{p\in P}\min_{q\in Q}\|p-q\|_2 + \frac{1}{|Q|}\sum_{q\in Q}\min_{p\in P}\|q-p\|_2.$$

The point-to-point term is L1 distance when $d(x, y) = \|x - y\|$ (the unsquared Euclidean norm used above), and L2 distance when $d(x, y) = \|x - y\|^2$ [5] – the squared variant used instead in this project's training loss (Section 5.6). The first term asks whether every deformed source point is near the target. The second term

asks whether every target point is covered by the deformed source. The symmetry is important: otherwise a small collapsed source could be close to part of the target while ignoring the rest.

Chamfer is powerful because it does not require point-to-point annotations. It is also limited because it is blind to semantic identity. It only asks whether surfaces are close in space. This project therefore implements local Chamfer terms computed separately inside PartField buckets combined with global Chamfer.

3.3 Why global Chamfer can be ambiguous

Chamfer distance is often used because it is simple, differentiable almost everywhere, and does not require the source and target meshes to have the same number of vertices. However, it treats both surfaces as unordered samples. If two regions are spatially near, Chamfer has no reason to prefer the semantically correct one. This creates three common ambiguities.

1. Part swapping: If two legs are close, the optimizer may exchange them. For a visual rendering this may be difficult to notice, but for correspondence it is a real error.
2. Part stealing: A large source region, usually the body, can cover target regions that should belong to a smaller part, such as a neck, ear, or tail.
3. Collapse: A source surface can shrink toward a high-density target region and still reduce one side of the nearest-neighbor loss.

The symmetric Chamfer formula reduces collapse because it also asks the source to cover the target, but it still does not know which region should cover which target part. Local PartField Chamfer is introduced exactly to reduce this freedom.

3.4 Surface distance versus surface correspondence

Surface distance and correspondence measure related but different properties. Surface distance asks whether the deformed shape occupies the same space as the target. Correspondence asks whether a source point moved to the correct target point or target region. This distinction is central to interpreting the experiments. A deformation can have excellent surface distance if the final mesh looks like the target, even when the mapping from source body parts to target body parts is partially wrong. In contrast, a deformation can preserve correspondence well while still having high surface distance if the source topology cannot represent the target geometry.

DenseCorr3D is useful because it supplies an independent semantic evaluator. The PartField losses are unsupervised and local; DenseCorr3D annotations are held out and later used to ask whether those unsupervised parts also agree with benchmark semantic regions. This separation is one of the strongest aspects of the evaluation design, and it is exactly what makes the comparison in Section 7.

4. Background and Related Work

4.1 Template fitting and mesh deformation

Classical mesh deformation often begins with a template mesh and tries to fit it to an observed target. The target may be a scan, a point cloud, a depth reconstruction, or another mesh. Because the template connectivity is preserved, the output remains easy to edit and compare. The difficulty is that the optimization must balance flexibility with regularity. A very flexible deformation can match the target surface but destroy the mesh. A very rigid deformation preserves the mesh but cannot adapt to new shape proportions.

This project follows the template-fitting spirit: the source mesh acts as the template, and the target mesh provides the geometric destination. The regularization terms keep the deformation from becoming arbitrary, while the PartField terms give it semantic structure. The method therefore lies between pure geometric registration and supervised correspondence learning.

4.2 Dense correspondence and DenseMatcher

Dense 3D correspondence methods aim to predict a mapping between surfaces. DenseCorr3D is a benchmark of meshes with annotated semantic groups, released alongside DenseMatcher [6], a learned method that predicts point-to-point correspondence directly from category-level features – it does not deform a mesh at all. Its goal is therefore not identical to the goal of this project. Still, the DenseCorr3D metrics are valuable because they answer an important question about the deformation: after the source has moved, did its semantic regions land on the correct target regions? DenseMatcher's own predicted correspondences were made available and can be scored on identical pairs with the identical evaluator used for this project's own numbers – turning "compared against a published number" into "compared against the method itself." Section 7 is dedicated to that comparison and its caveats.

4.3 Unsupervised part discovery

Part discovery is the task of decomposing a shape into meaningful regions. In a supervised setting, a dataset might label vertices as head, body, leg, handle, blade, seat, back, and so on. In this project, that kind of supervision is not assumed. PartField provides a practical alternative by producing feature fields that can be clustered into buckets. These buckets are not named semantic classes, but they often align with repeated object structure.

The main risk is that unsupervised buckets can be inconsistent. If source bucket 3 is a leg but target bucket 3 is partly body, the local loss becomes misleading. This is why the project tests multiple granularities and why alignment is so important. PartField is not treated as an oracle; it is treated as a useful but imperfect prior.

4.4 Differentiable geometry losses

Modern mesh optimization often uses differentiable geometric losses. Chamfer distance, normal- consistency losses, edge-length penalties, and Jacobian-style regularizers [10, 11] can all be optimized with gradient-based methods. The project uses this family of ideas, but the key addition is the local semantic decomposition. Instead of applying one global nearest-neighbor loss to all samples, the optimizer also receives local nearest-neighbor losses inside corresponding buckets. This small change has a large effect on the behavior of the optimization.

5. Pipeline Overview

The pipeline takes a source mesh and a target mesh as input. The source is the object that will be deformed. The target is the object whose shape should be matched. The method keeps the source topology: vertices move, but the original face connectivity remains. This makes the result easy to compare with the original source, but it also means the method cannot create a new limb or a new disconnected component from nothing.

5.1 System overview: bucket selection, segmentation, deformation

At the highest level the system has three stages (Figure 2): a *bucket selector* decides how many PartField buckets K to use for a given source/target pair; *PartField* co-segments both meshes into K buckets; and the *deformation* stage optimizes the source toward the target under Chamfer-style guidance plus regularization. The implemented objective supports both whole-surface and per-bucket terms.

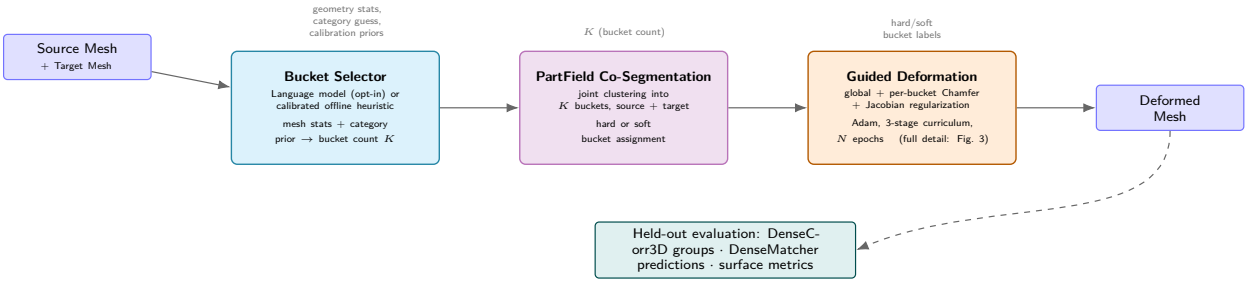


Figure 2. Bird's-eye system overview. A bucket selector, implemented either with an LLM prompt or with the offline heuristic, turns mesh geometry plus project/user priors into a single PartField bucket count K. PartField then co-segments the pair, and deformation consumes those buckets as in Figure 3.

Earlier experiments in this report chose PartField granularity by hand or by sweeping a fixed range per category (Section 7); the selector automates that choice for a specific source/target pair using an LLM prompt. It parses both meshes directly (vertex and face counts, bounding-box aspect ratio, connected-component count), guesses an object category from the mesh paths, and can receive short user notes about what has worked or failed in previous DenseCorr3D/DenseMatcher experiments. The LLM-backed selector is therefore not a blind zero-shot selector: it combines mesh statistics, the user's qualitative prior for the category, and general object knowledge (for example, one or two buckets for smooth single-part produce; more buckets for articulated animals, chairs, cars, or tools). This component was built mainly so that the pipeline has one automated, complete path from a raw source/target pair to a bucket count, rather than requiring a hand-chosen K for every new pair.

A concrete example is `horse2→giraffe`. The selector recognized the pair as an animal case from the mesh paths and general object knowledge, and returned $K = 6$ with confidence 0.78; a PartField co-segmentation preflight then validated that bucket count before handing the pair to the deformation stage.

5.2 The deformation stage in detail

Figure 3 expands the PartField and deformation stages of Figure 2 into the full mechanism: the semantic-extraction chain, the three loss terms and their combination, the curriculum, and the evaluation branch.

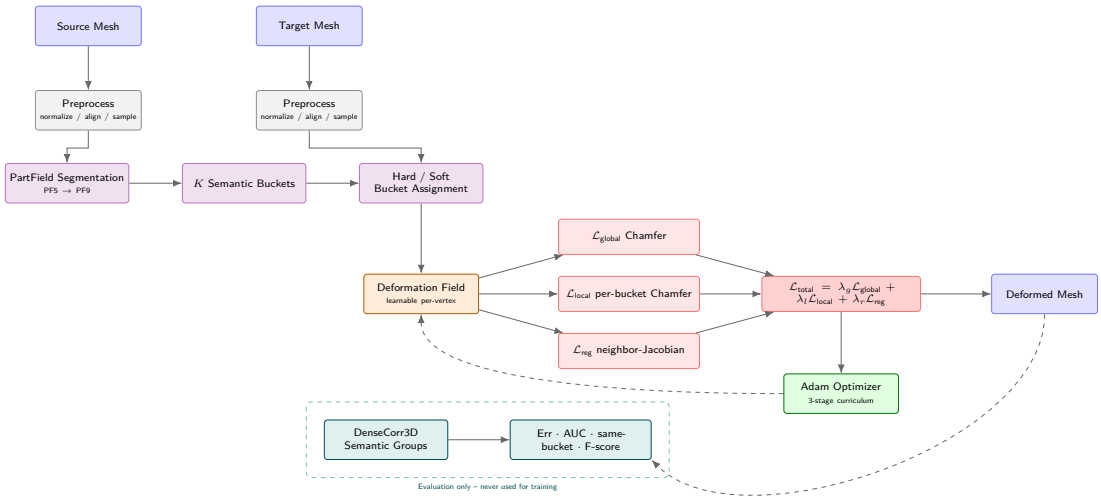


Figure 3. Detailed PartField segmentation and guided-deformation stages.

Figure 3 shows the full process. At a high level:

1. Normalize and orient the source and target meshes.
2. Optionally, ask the bucket selector (Section 5.1) to choose the PartField bucket count K for this pair from mesh geometry and calibration priors; otherwise K is set by hand.
3. Extract PartField features and cluster them into K part buckets.
4. Build hard or soft bucket assignments for vertices and faces.
5. Optimize a deformation field using the active Chamfer terms for that experiment family and regularization.
6. Evaluate geometry on sampled surfaces and semantic alignment on held-out DenseCorr3D groups, and (Section 7) against DenseMatcher's own predictions, when available.

The mesh files provide vertices and faces; PartField inference produces features; clustering produces label files; the deformation run produces a final mesh and intermediate visualizations; and the evaluation scripts produce summaries. This makes the pipeline easy to debug because each stage can be inspected separately. If a final deformation fails, one can ask: was the mesh aligned correctly, did PartField produce reasonable buckets, did the optimizer move the right regions, or did the metric penalize a thin semantic group?

Object	Meaning in the pipeline	Common failure if handled badly
Source mesh	The topology that will be preserved and deformed.	Wrong source scale or pose creates unnecessary distortion.
Target mesh	The geometry the source should match.	Missing components or bad orientation mislead Chamfer and PartField.
PartField features	Per-point or per-region descriptors used for bucket construction.	Features can cluster by pose or noise rather than semantic part.
Bucket labels	Local regions used by the semantic Chamfer loss.	Bucket mismatch causes semantically wrong local pulls.
Final mesh	The source topology after deformation.	Can fit globally while still failing local correspondence.

Table 1. The main intermediate objects in the pipeline and why each one matters.

5.3 Source and target preparation

All meshes are converted into the same coordinate convention before optimization. The pipeline normalizes the source and target into a unit box, optionally rotates or aligns the mesh, and samples points on the surface for loss evaluation. This step is simple but critical. If two animals face opposite directions, or if a vegetable category has inconsistent stem orientation, PartField buckets can be swapped even before deformation begins. Several failure cases in the project were eventually traced to alignment rather than to the optimizer itself.

For DenseCorr3D experiments, the meshes come with semantic groups and correspondence annotations. These annotations are not used in the deformation. They are held out for evaluation, where they provide an independent way to ask whether a deformed source part lands on the correct target part. For SAM3D experiments, the meshes are reconstructed from real photographs and are evaluated mainly by surface geometry and PartField-region agreement, since no independent ground-truth correspondence exists for self-created meshes.

5.4 PartField buckets

Figure 4 shows what PartField actually does to a pair of meshes before the rest of this section works through the mechanics.



Figure 4. Giraffe-to-elephant PartField decomposition, PF7–PF10.

PartField [7] maps a mesh to local features that can be clustered into repeated structural regions. The available granularities are written as PF1 through PF10 (a lower number creates coarser buckets; a higher number creates finer buckets). Coarse buckets are more stable but can merge distinct parts. Fine buckets can separate legs, head, body, ears, or tail more clearly, but they can also become unstable when the source and target have different topology or pose.

The important design choice is that the same bucket index is used on both source and target after joint clustering or bucket balancing. Concretely, *joint clustering* means the source and target meshes' PartField feature vectors are pooled into one combined array before clustering, and a single K-means model (with K equal to the active bucket count) is fit on that pooled set; splitting the resulting per-point cluster assignments back out by mesh is what gives source and target their shared bucket indices, with no separate cross-mesh matching step required afterward – bucket k means the same region of PartField feature space on both meshes because both meshes' points were clustered together in the first place. Bucket k on the source is therefore encouraged to move near bucket k on the target. The buckets are not hand-labeled. The method never needs an explicit label such as "front left leg". It only needs a consistent decomposition into local regions.

The project also distinguishes raw buckets from bucket-balanced buckets. Raw clustering can produce a dominant bucket that covers most of the body and small buckets that contain only a few vertices. This is a problem for local losses: a very large bucket can behave like another global Chamfer term, while tiny buckets become noisy. Bucket balancing makes the local supervision more usable by preventing one region from overwhelming the optimization, as shown concretely in Figure 5.

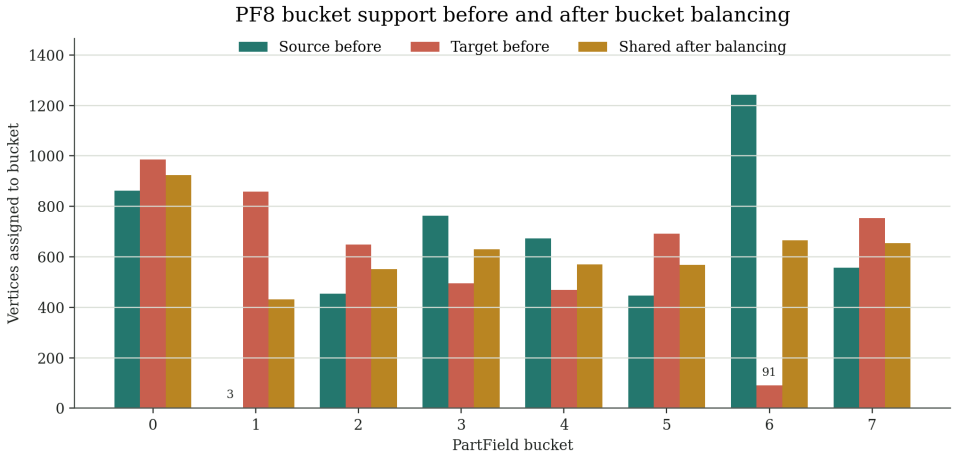


Figure 5. PF8 bucket support before and after bucket balancing.

5.5 Deformation representation

The deformation is parameterized as a learnable *per-face Jacobian* field rather than raw vertex positions: a 3×3 Jacobian is optimized for every face of the source mesh (via a `NeuralJacobianFields.SourceMesh` object [11]), and vertex positions are reconstructed from the current Jacobians at every optimization step by a Poisson-style solve. This project uses the Jacobian parameterization, optimized directly by Adam at learning rate 0.0025. The neighbor-Jacobian regularizer penalizes large differences between the Jacobians of adjacent faces, which is what keeps the surface from folding or self-intersecting as it deforms; the reconstruction step is what turns a smooth Jacobian field into a smooth vertex displacement.

Keeping the source topology has an important consequence: the result implicitly defines a correspondence from the original source to the final deformed mesh. Vertex i before optimization is still vertex i after optimization; only its position changes. This is useful for evaluation and visualization because colors, labels, and source regions can be propagated directly (it is also what makes the color-consistency fix behind Figure 21 possible – see its caption). It is also why over-deformation is dangerous: if a group of source vertices is pulled across the object to satisfy Chamfer, the mesh may still look near the target, but the inherited source labels reveal that the semantic mapping is wrong.

5.6 Loss function

The optimization objective is

$$\mathcal{L}_{\text{total}} = \lambda_g \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{local}} + \mathcal{L}_{\text{reg}}.$$

The global term $\mathcal{L}_{\text{global}}$ is the Chamfer distance [9] between samples from the entire deformed source surface and samples from the target surface. The local term is a sum of *asymmetric* Chamfer distances over active PartField buckets (buckets with at least 12 vertices on both sides by default):

$$\mathcal{L}_{\text{local}} = \sum_{k \in \mathcal{K}_{\text{active}}} d^{\text{asym}}_{\text{Chamfer}}(P_k, Q_k),$$

where P_k are samples from the deformed source bucket k and Q_k are samples from the target bucket k. Concretely, for $p \in P_k$ write $q^*(p) = \arg\min_{q \in Q_k} \|p - q\|_2$ for its nearest target neighbor, and for $q \in Q_k$ write $p^*(q) = \arg\min_{p \in P_k} \|q - p\|_2$ for its nearest source neighbor. Call p *claimed* if $p = p^*(q)$ for at least one $q \in Q_k$ (some target point in the bucket needs exactly this source point), giving it weight $w(p) = 1$; an unclaimed source point instead gets weight $w(p) = \gamma < 1$. Then

$$d^{\text{asym}}_{\text{Chamfer}}(P_k, Q_k) = \alpha \cdot \frac{\sum_{p \in P_k} w(p)\, \|p - q^*(p)\|_2^2}{\sum_{p \in P_k} w(p)} \;+\; \beta \cdot \frac{1}{|Q_k|} \sum_{q \in Q_k} \|q - p^*(q)\|_2^2,$$

with source-to-target weight $\alpha \in [0.30, 0.40]$ (best-performing presets use $\alpha = 0.35$, denoted `asym035` in job names) and target-to-source weight $\beta = 1.0$; the unmatched-point down-weight γ (tested at $\gamma = 0.15$, the `mutual015` preset) is only active when explicitly enabled. Unlike the global Chamfer distance in Section 3.2 (raw $\|\cdot\|_2$, for interpretability in real units), the local term is computed on squared distances, the standard choice for a smoother optimization signal; only the reporting metrics in Section 5.7 use the raw distance.

Both mechanisms exist to counter the same failure mode. Under a plain, evenly-weighted per-point Chamfer term, a source point that no target point in the bucket actually needs (i.e. unclaimed) is still pulled at full strength toward its own single nearest target point; because nothing else constrains that point to stay near its neighbors, the result can be a thin spike or an over-elongated limb rather than a smooth deformation. Lowering α reduces how hard every source point is pulled toward the target on its own, and down-weighting unclaimed points removes most of the pull specifically for the points most likely to spike. Figure 25 in Appendix D shows exactly this failure on an early bulldog→dachshund run that used the chamfer-only objective before this weighting was introduced. This is the main semantic guidance in the method. In the headline DenseMatcher comparison (Section 7), the saved run configurations set $\lambda_g = 0$ and use the PartField/local term as the active geometric fitting signal; global-only and global+PartField variants are therefore not folded into those headline numbers.

The regularization term discourages unrealistic deformations:

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{jn}}\mathcal{L}_{\text{neighbor-Jacobian}} + \lambda_{\text{jump}}\mathcal{L}_{\text{edge-jump}} + \lambda_{\text{es}}\mathcal{L}_{\text{edge-stretch}}.$$

The neighbor-Jacobian term penalizes differences between the Jacobians of adjacent faces, acting directly on the deformation's own parameterization (Section 5.5). The edge-displacement-jump term penalizes cases where the displacement difference between two edge-connected vertices exceeds a threshold fraction of the original edge length, with an extra penalty on the single worst violation – this is the term most directly responsible for preventing self-crossing in thin, elongated parts such as necks and legs.

Loss component	What it measures	Why it is needed
Global Chamfer	Distance between the whole deformed source surface and the whole target surface.	Gives the optimizer the overall target shape.
Local Chamfer	Asymmetric distance between matching PartField buckets.	Reduces part swapping and part stealing.
Bucket weights	Relative influence of each bucket.	Keeps small parts from disappearing under large body regions.
Neighbor-Jacobian	Agreement between nearby per-face Jacobians.	Prevents abrupt discontinuities and folding across adjacent faces.
Edge stretch / jump penalties	Changes in edge length or displacement across edges.	Discourages folds, spikes, and extreme local stretching.

Table 2. Interpretation of the main loss components.

5.7 Evaluation metrics

The report uses several metrics, defined here using the same P (deformed source samples) and Q (target samples) notation as the Chamfer distance in Section 5.1.

F@0.02 is the harmonic mean of precision and recall at a unit-box threshold $\tau = 0.02$:

$$\text{Prec}(\tau) = \frac{1}{|P|}\sum_{p\in P} \mathbb{1}\left[\min_{q\in Q}\|p-q\|_2 < \tau\right], \qquad \text{Rec}(\tau) = \frac{1}{|Q|}\sum_{q\in Q} \mathbb{1}\left[\min_{p\in P}\|q-p\|_2 < \tau\right],$$

$$\text{F@}\tau = \frac{2\,\text{Prec}(\tau)\,\text{Rec}(\tau)}{\text{Prec}(\tau)+\text{Rec}(\tau)}.$$

Precision asks what fraction of deformed-source surface samples lie within τ of the target; recall asks what fraction of target samples are covered by the deformed source. Higher is better.

The robust $p95$ distance is the 95th percentile of the same per-point nearest-neighbor distances used by Chamfer, $p95 = Q_{0.95}\big(\{\min_{q\in Q} \|p-q\|_2 : p \in P\}\big)$, where $Q_{0.95}$ denotes the 95th-percentile operator. It is useful because a method can have good mean Chamfer while a small part of the mesh is very wrong. Lower $p95$ means fewer large local errors.

DenseCorr3D-style correspondence evaluation uses normalized geodesic error. DenseMatcher defines the semantic-group version of this metric as the distance from the predicted target point to the nearest point in the correct ground-truth semantic group [6]. For a source point i with predicted target correspondence $\hat{y}_i$ and target semantic group G_i^*,

$$e_i = \frac{\min_{y\in G_i^*} d_{\text{geo}}(\hat{y}_i, y)}{\sqrt{\text{area(target)}}},$$

where d_{geo} is graph-geodesic (shortest-path) distance on the target mesh and the $\sqrt{\text{area}}$ term normalizes for object scale. $\hat{y}_i$ means something different for each method being scored this way. DenseMatcher predicts $\hat{y}_i$ directly: it is the point its `surface_map`/`surface_map_inv` output assigns to source point i. This pipeline does not output a correspondence map at all (Section 5.1), so $\hat{y}_i$ is instead induced from its deformation output after the fact: source vertex i keeps its identity through optimization (Section 5.5), so $\hat{y}_i$ is defined as the nearest target-mesh vertex, by Euclidean distance in the same normalized coordinates used throughout this section, to vertex i's final deformed position. This is the same nearest-neighbor lookup already used by every Chamfer-based term and metric in this report; evaluation simply reuses it to name a correspondence rather than only a distance. The report follows the evaluator output and reports

$$\text{Err} = 100 \cdot \frac{1}{n}\sum_i e_i,$$

so an Err value of 4.77 means mean normalized semantic geodesic error 0.0477. Let $\Phi(t) = \frac{1}{n}\sum_i \mathbb{1}[e_i < t]$ be the fraction of points correct at threshold t; AUC integrates this curve up to a normalized threshold of 0.25 and rescales it to $[0, 1]$:

$$\text{AUC} = \frac{1}{0.25}\int_0^{0.25} \Phi(t)\, dt.$$

For AUC, higher is better; for Err, lower is better.

The remaining reported metrics are surface diagnostics. Symmetric Chamfer is the mean nearest-neighbor surface distance in both directions in unit-box coordinates. Normal consistency is the mean absolute dot product between matched surface normals, so higher values indicate smoother normal agreement. Semantic Chamfer and semantic F@0.02 repeat the surface metrics inside matching semantic or PartField regions rather than over the whole object. Same-bucket agreement is the fraction of evaluated points whose predicted target neighbor falls in the expected bucket or semantic group; it is useful as a coarse label-consistency check but does not replace geodesic AUC/Err.

5.8 Runtime and computational cost

All deformation runs use a single GPU on a shared Slurm cluster. Table 3 compares the currently submitted resource request (`gpu:1`, 8 CPUs, 10 GB RAM, 30-minute limit) against measured usage: peak RAM stays below 1.8 GiB and wall-clock time ranges from about 4 to 18 minutes across job families – a small fraction of the 80,000+ diffusion evaluations and 40–64 GB requests the earlier MeshUp/SDS branch needed (Appendix B).

Resource	Currently submitted	Actual measured need	Recommended request
GPU	1 × H100	1 GPU (single-GPU pipeline)	1 GPU
CPUs	8	∼0.7–1.0 core average	2 (4 to be safe)
RAM	10 GB	max 1.8 GiB system RAM	4 GB (8 GB to be safe)
Time limit	30 min	∼4–18 min	20–30 min (safe margin)

Table 3. Slurm resource request vs. measured usage, standard deformation jobs.

DenseMatcher's paper reports that computing one pair's correspondence takes 8.4–12.4 seconds on an A100 GPU, a single feed-forward pass with no iterative optimization [6]. This pipeline takes 4–18 minutes per pair (Table 3) on an H100, since it solves a per-pair iterative optimization problem instead of one forward pass.

5.9 Reproducibility and held-out evaluation

The implementation is organized around repeatable experiment families. Each run stores its configuration, final mesh, visualizations, and evaluation outputs. This matters because mesh deformation experiments are sensitive to preprocessing, PartField level, target choice, and loss weights.

DenseCorr3D's semantic groups are used only to score a deformation after the fact. They are never read by the deformation stage, the loss function, or the bucket selector: the only signal PartField and the optimizer ever see is geometry. This separation is what makes the DenseCorr3D-based numbers in this report an external check on the result.

6. Data and Assets

6.1 DenseCorr3D

DenseCorr3D is a dense 3D correspondence benchmark with meshes, semantic groups, and evaluation tools for surface correspondence, released alongside DenseMatcher [6]. In this project, its 23 categories – animals, produce, tools, vehicles, and containers – provide an external test bed for semantic deformation: after deforming a source mesh into a target mesh, the DenseCorr3D annotations measure whether corresponding surface regions land in the right place, and the breadth of categories is what lets the report ask where PartField-guided deformation generalizes and where it fails.

6.2 SAM3D and project meshes

The project also uses a self-created set of meshes reconstructed with SAM3D [8] from single real photographs (each paired with a SAM2-derived mask) and then processed into the same mesh deformation pipeline. These assets are useful because they test the method on realistic meshes outside the benchmark distribution: imperfect components, noisy reconstructions, pose differences, and inconsistent scale that curated benchmark meshes do not have.

Dataset / split	Scale	Role in the report	Main caveat
DenseCorr3D categories	23 categories; 2328 evaluated (directed pair, PF-setting) rows	Main semantic alignment benchmark against held-out DenseCorr3D groups and DenseMatcher's released predictions.	Graph-geodesic local evaluator; PF ranges are category-specific and range-averaged.
SAM3D-created animals	14 meshes; 8 directed pairs evaluated	Self-created reconstructed animal set for surface-quality evaluation and video showcase.	No independent dense correspondence ground truth.
SAM3D-created vehicles	25 meshes; 5 directed pairs evaluated	Rigid, sharp-edged non-animal test outside DenseCorr3D.	Markedly weaker than animals under the identical PartField-only recipe.
Cross-dataset chain examples	2 directed pairs	Transfer between benchmark-like and reconstructed assets.	Small qualitative and diagnostic set.

Table 4. Combined dataset inventory and evaluation role.

7. Fair Comparison Against DenseMatcher Under Graph-Geodesic Evaluation

This is the central quantitative section of the report. DenseMatcher's own predicted correspondences (`surface_map` / `surface_map_inv`, DenseMatcher's method-0 forward functional-map output) were computed for the DenseCorr3D test split. Both methods are scored with the identical graph-geodesic evaluator – Dijkstra shortest paths on the target mesh, since the official DenseCorr3D geodesic distance matrices are not available – on the identical directed pairs, and the result is reported with its evaluation caveats. Everywhere in this section, "beats DenseMatcher" means beats DenseMatcher's released predictions under this graph-geodesic evaluation protocol.

7.1 Methodology

Both methods are scored against the identical DenseCorr3D `groups.txt` semantic labels, using the identical graph-geodesic (Dijkstra) distance, identical $\sqrt{\text{area}}$ normalization, and identical AUC$[0, 0.25]$/Err formula, where Err is mean normalized semantic geodesic error multiplied by 100 (Section 5.7). DenseMatcher's number in every table below is restricted to the precise (source, target) directed pairs this pipeline was evaluated on, and direction was matched explicitly (`surface_map` vs. `surface_map_inv`) per pair. DenseMatcher's predictions never touched the deformation pipeline: the only active Chamfer term in every run reported here is the PartField/local term, used together with the same regularizers described in Section 5.5 as in every other experiment family. DenseMatcher's `.npz` files only enter the project in a separate evaluation script, after every deformation run already existed.

DenseMatcher's predictions were computed on DenseCorr3D's released meshes and were never regenerated by this project: the `surface_map`/`surface_map_inv .npz` files were supplied precomputed (Section 6) and read directly by the evaluation script, with none of this pipeline's own mesh preprocessing – normalization, rotation, or alignment – ever applied to the meshes DenseMatcher saw. This pipeline's own input meshes, by contrast, were sometimes reoriented before deformation: Table 6's "Mesh prep" column records this per category. This is an asymmetry worth stating plainly rather than leaving implicit: for categories marked "assisted" in that table, this pipeline received a canonicalization step before PartField ever ran that DenseMatcher's released predictions never had access to. The clean-preparation rows reported throughout this section exist specifically to give a reading of the comparison with that asymmetry removed; the all-categories rows should be read with it in mind.

DenseCorr3D's official evaluator and its per-mesh geodesic distance matrices were never published, so this report does not attempt to reproduce DenseMatcher's officially published benchmark result. For calibration,

that published result is AUC 0.845/Err 1.74 on DenseMatcher's full DenseCorr3D test split and AUC 0.775/Err 2.82 on its held-out split, under DenseMatcher's own exact-geodesic evaluator [6] – a different evaluator, and for some categories a different mesh orientation, than the comparison below uses, so it is context rather than a number this report's numbers should be expected to match. Instead, a graph-geodesic (Dijkstra) evaluator was implemented and applied identically to both methods, using DenseMatcher's own released `surface_map` predictions directly (a couple of alternative evaluator designs were also tried and are noted briefly in Appendix G). The claim here is narrower and cleaner: on identical directed pairs, using the same graph-geodesic scorer for both outputs, the PartField-guided deformation pipeline beats DenseMatcher in 21 of 23 categories.

The term *zero PF selection* in this section does not mean that every category was swept over PF01–PF10. It means that, once a category-specific PF range was declared, the report averages every tested granularity in that range and does not choose the best-scoring PF level for the headline. The PF ranges differ by category because they were chosen from a simple part-complexity prior before aggregation: categories expected to have more repeated or articulated parts, such as animals, backpack, cars, chairs, and shampoo, were assigned broader or higher ranges; smoother produce categories, such as banana and egg, started from lower ranges; and uncertain produce cases with stems, ridges, or ambiguous internal regions were swept more broadly. The one place a best-per-category number appears at all is Table 5's "best PF per category" row, and it is always labeled explicitly as an upper bound rather than mixed into the zero-selection numbers.

A first pass at this comparison found that 22 of 23 categories beat DenseMatcher. A margin that wide calls for verification before being reported as a final number, so the result was put through a verification audit: checking specifically for sample-selection bias (none found – DenseMatcher scores marginally *better*, not worse, on the matched subset than on its own full unrestricted set) and for direction/index bugs (none found).

7.2 Main results

				Pipeline		DenseMatcher	
Category subset	PF protocol	Categories	Wins	AUC	Err	AUC	Err
All categories	Range-averaged	23	21/23	0.824	4.77	0.672	11.69
Clean-preparation	Range-averaged	12	10/12	0.806	5.52	0.713	9.65
All categories	Best PF per category	23	22/23	0.854	3.74	0.672	11.68
Clean-preparation	Best PF per category	12	11/12	0.851	3.89	0.714	9.65

Table 5. Headline graph-geodesic comparison against DenseMatcher predictions, split by category subset and PartField (PF) selection protocol. "Category subset" is either all 23 available categories or the clean-preparation subset of 12 categories whose meshes did not need assisted orientation or reconstruction. "PF protocol" is either range-averaged (the zero-best-PF protocol: every tested granularity in the predeclared category range is averaged, rather than reporting only the best setting) or best PF per category, an oracle upper bound shown for context and not part of the headline claim.

Across all 23 available categories, and without choosing a PartField granularity to win, this pipeline reaches AUC 0.824/Err 4.77 against DenseMatcher's AUC 0.672/Err 11.69, winning 21 of 23 (Table 5). When assisted preparation categories are removed, the pipeline still wins 10 of 12 categories with the same zero-best-PF protocol. This matters because PartField is useful but not rotation-invariant; as Figures 18 and 19 show, source and target meshes need a comparable canonical direction for the unsupervised buckets to line up cleanly. The result is therefore strongest when read together with a practical requirement: use simple orientation normalization before PartField. The two best-PF rows are shown for context, not as the primary claim: letting the pipeline pick its best PartField granularity per category – an oracle upper bound no real deployment would have access to – raises the all-category win count to 22 of 23 and the clean-preparation win count to 11 of 12. Eggplant's loss under range averaging is a granularity problem and disappears once the best setting is picked; broccoli's is not, and it remains the one category this pipeline loses even under oracle PartField selection, for the structural reasons traced in Section 9.1.

This comparison should not be read as a universal claim that deformation beats an already aligned source

mesh. The claim defended here is narrower: under the same local graph-geodesic scorer, the PartField-guided outputs outperform DenseMatcher's released predictions on the matched directed pairs.

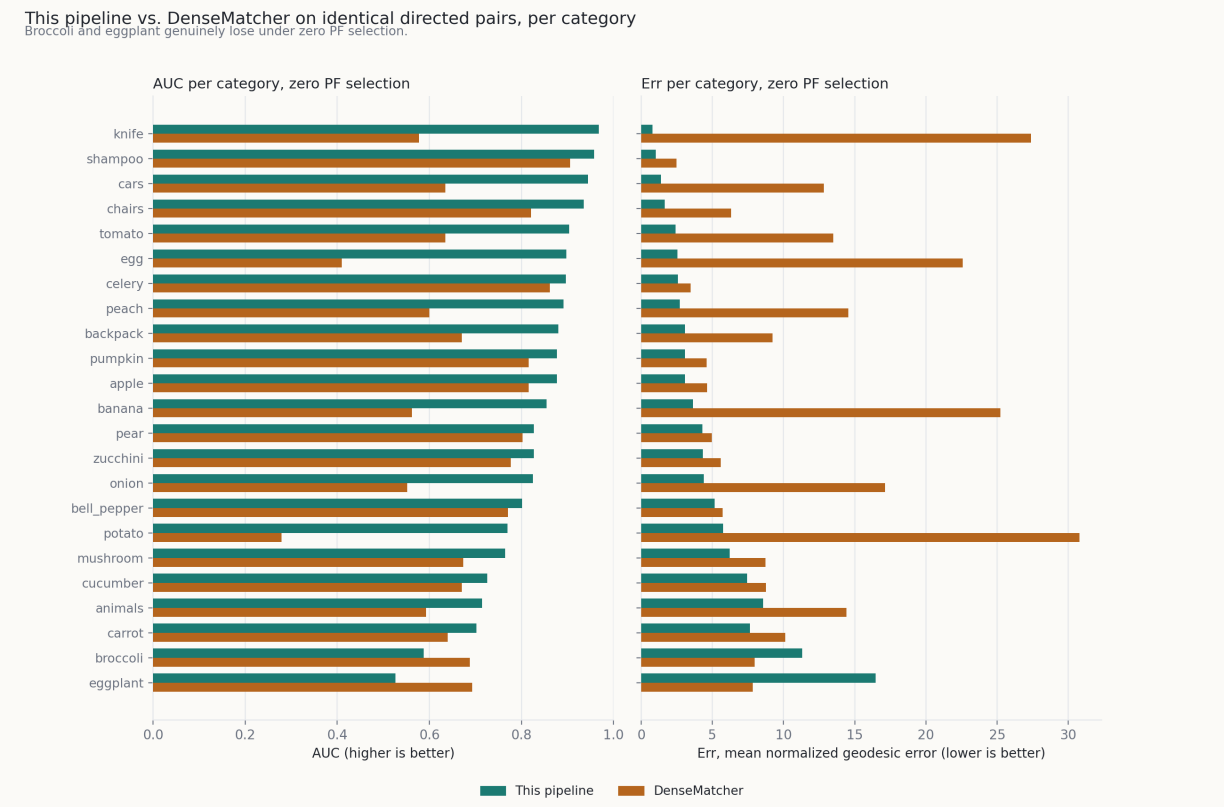


Figure 6. Per-category AUC and Err under range-averaged, zero-best-PF selection. Broccoli and eggplant lose.

7.3 Qualitative comparison: on the mesh

The numbers above compare correspondence quality; DenseMatcher itself only ever predicts a per-vertex point correspondence (a functional map over DiffusionNet features refining multi-view SD-DINO projections) [6] and does not output a deformed mesh. To make it visually comparable to this pipeline's output, the mesh that DenseMatcher's own predicted correspondence implies gets rendered: the source mesh, unchanged in topology, with every vertex moved to the position of its predicted match on the target – exactly the lookup already used to score AUC/Err, rendered here instead of only scored. Figure 7 shows this for four clean-provenance pairs spanning the range of DenseMatcher failure severity actually measured: knife (near-total failure, AUC 0.219), potato (a severe failure, AUC 0.357), cars (a milder failure, AUC 0.469), and shampoo (the mildest gap of the four, AUC 0.769). Source vertices carry a fixed position-based rainbow; Final and the DenseMatcher warp inherit it unchanged, since both share the source mesh's vertex indices. A spatially coherent result keeps the rainbow smooth; a confused correspondence scrambles it – the standard texture-transfer diagnostic from the shape-correspondence literature, applied here to both methods on equal footing.

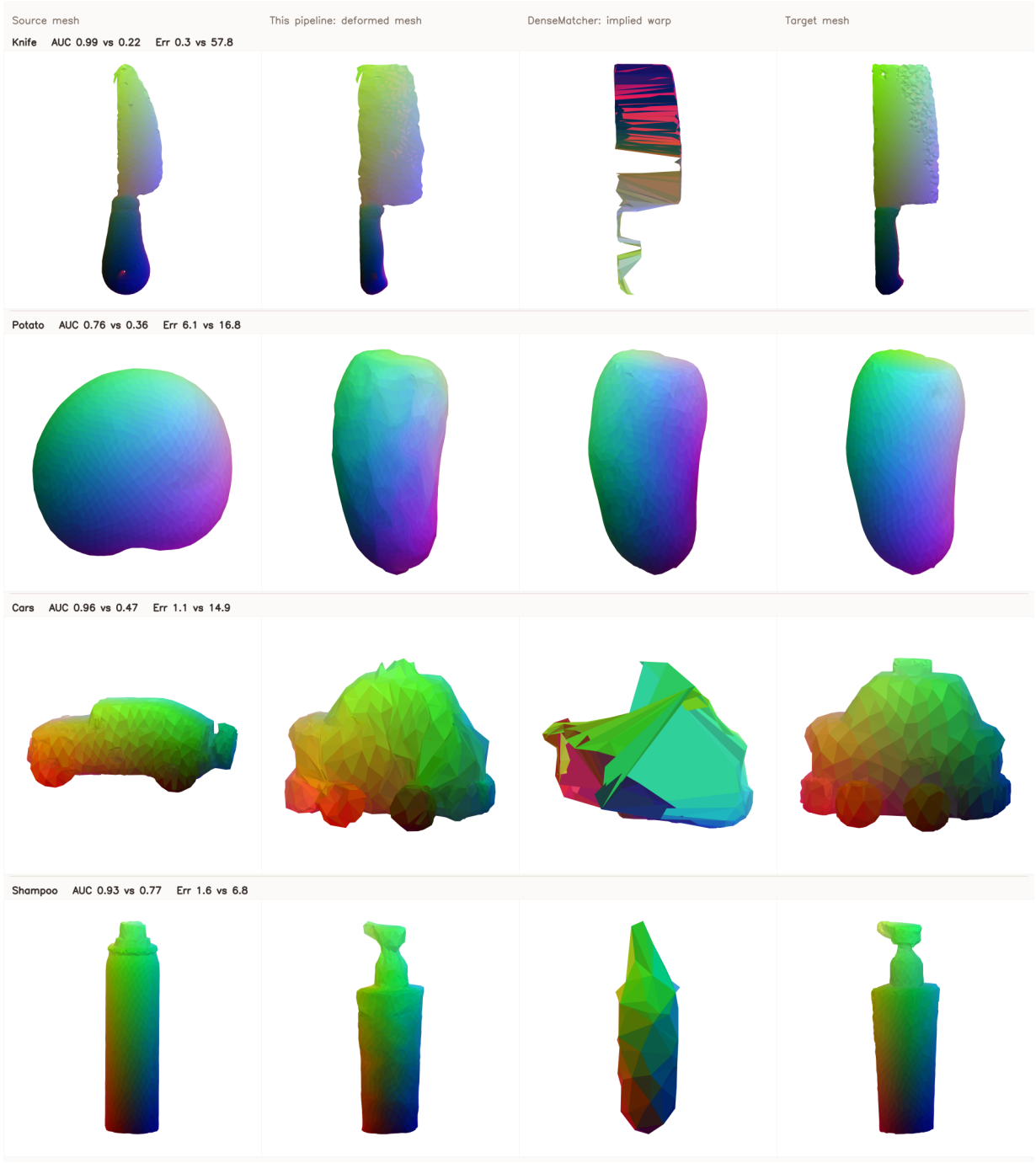


Figure 7. Source, Final, DenseMatcher's implied warp, and target, for four clean-provenance pairs. Both warps inherit the source's rainbow coloring, so scrambling marks incoherence.

Figure 7's knife, cars, and shampoo rows visibly fold and self-intersect under DenseMatcher's implied warp. Potato is the exception: despite the largest AUC gap of the four, its warp stays smooth and blob-like – less accurate, but not torn. This gap between a summary statistic and the rendered geometry has a concrete explanation: AUC/Err integrate partial credit over a normalized-geodesic threshold sweep, so a correspondence that is uniformly somewhat wrong can post a similar AUC hit to one that is catastrophically wrong on a smaller fraction of vertices, even though only the latter crushes the surface. Potato is the former case; knife,

cars, and shampoo are the latter. Figure 8 illustrates the mechanism behind that gap – it is a diagnostic for the many-to-one failure explained below, on four pairs chosen to span DenseMatcher's failure severity rather than to represent typical performance, not a general quantitative benchmark – with a topology-preserving measure unrelated to AUC/Err: for every face, the median $|\log_2(\text{deformed edge length}/\text{source edge length})|$, comparing the same source edges under this pipeline's Final mesh and under DenseMatcher's implied warp. A value of 0 means a face moved rigidly; large values mean it was locally crushed or torn. On three of the four pairs, DenseMatcher's implied warp is torn by two to nearly eight orders of magnitude more than this pipeline's output; on potato, the two are comparable, confirming that its failure is a smooth but systematically wrong correspondence rather than a torn one.

The concrete mechanism behind that tearing is a many-to-one correspondence: nothing forces DenseMatcher's predicted map to be one-to-one, so several different source vertices are free to land on the same (or a nearly identical) target vertex. Every edge between two such vertices then collapses toward zero length, which is exactly what a large edge-distortion score is picking up. This pipeline cannot fail this way by construction: it moves the source mesh's own vertices directly, so its output always keeps the source's vertex count and connectivity, with no correspondence step that could collapse several vertices onto one point.

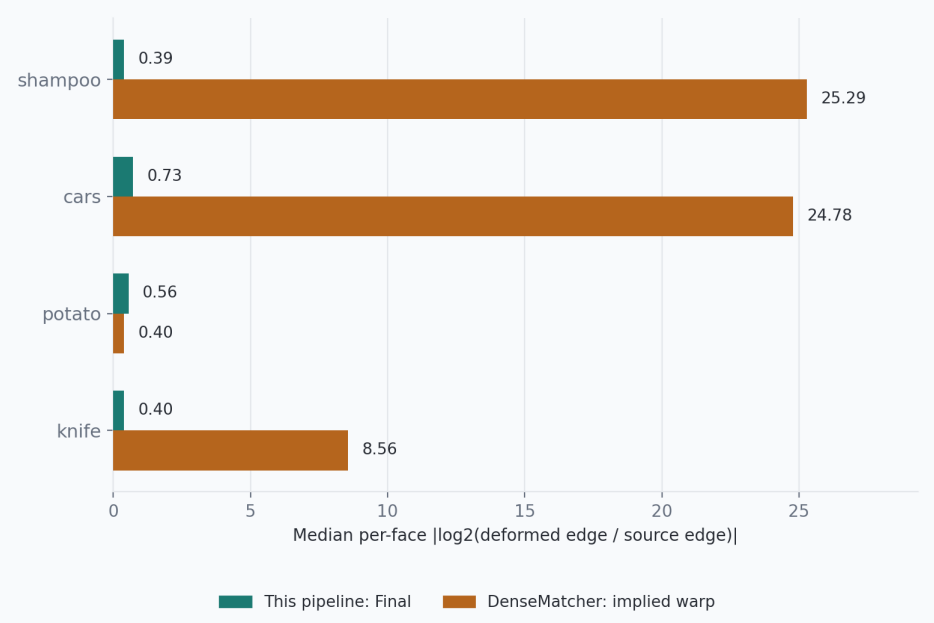


Figure 8. Median per-face local edge-stretch distortion for the same four pairs (linear scale), illustrating the many-to-one correspondence failure described in the text rather than serving as a general quantitative benchmark: these four pairs were chosen to span DenseMatcher's failure severity, not sampled to represent typical performance. Lower is better.

7.4 Full per-category table

Table 6 also records input preparation. "Clean" means the category used the canonical meshes directly or only label-free geometry/visual orientation checks. "Assisted" means additional preparation was needed to put source and target into comparable orientations before PartField was applied.

Table 6. Full per-category comparison under range-averaged, zero-best-PF selection. Rows are evaluated (directed pair, PF-setting) entries, not unique directed pairs.

Category	Mesh prep	Rows	PF range	Ours AUC	Ours Err	DM AUC	DM Err	Verdict
animals	clean	180	PF01–09	0.7148	8.578	0.5934	14.429	win
apple	assisted	80	PF05–10	0.8770	3.078	0.8154	4.620	win
backpack	clean	60	PF04–08	0.8805	3.063	0.6706	9.223	win
banana	assisted	120	PF01–04	0.8546	3.639	0.5618	25.220	win
bell_pepper	assisted	72	PF01–04	0.8018	5.170	0.7712	5.726	win

Category	Mesh prep	Rows	PF range	Ours AUC	Ours Err	DM AUC	DM Err	Verdict
broccoli	clean	120	PF01–06	0.5875	11.313	0.6883	7.978	**lose**
carrot	clean	36	PF02–04	0.7025	7.648	0.6404	10.132	win
cars	clean	100	PF04–08	0.9449	1.397	0.6345	12.843	win
celery	clean	45	PF01–06	0.8967	2.583	0.8615	3.464	win
chairs	clean	100	PF04–08	0.9350	1.658	0.8212	6.329	win
cucumber	clean	120	PF01–06	0.7255	7.426	0.6708	8.768	win
egg	assisted	150	PF01–05	0.8977	2.562	0.4095	22.588	win
eggplant	clean	36	PF01–06	0.5265	16.475	0.6927	7.847	**lose**
knife	clean	48	PF02–05	0.9679	0.802	0.5780	27.370	win
mushroom	assisted	120	PF01–04	0.7642	6.209	0.6734	8.724	win
onion	assisted	121	PF01–06	0.8246	4.401	0.5523	17.125	win
peach	assisted	120	PF01–05	0.8917	2.706	0.6000	14.559	win
pear	clean	120	PF01–04	0.8275	4.313	0.8027	4.953	win
potato	assisted	80	PF01–06	0.7701	5.754	0.2791	30.792	win
pumpkin	assisted	120	PF02–07	0.8771	3.073	0.8163	4.599	win
shampoo	clean	80	PF01–04	0.9584	1.043	0.9058	2.497	win
tomato	assisted	120	PF01–07	0.9038	2.404	0.6348	13.481	win
zucchini	assisted	180	PF01–06	0.8272	4.341	0.7767	5.588	win

7.5 Published DenseMatcher numbers, where available

DenseMatcher's paper does not publish a full 23-category breakdown; it reports only the aggregate numbers already given above, plus one appendix table comparing against a single baseline (URSSM) on four categories chosen for "complex topologies" [6]. Table 7 reproduces that appendix table's DenseMatcher column next to both the reimplemented DenseMatcher score from this report's own graph-geodesic evaluator and this pipeline's own score, for the four categories where a published per-category number exists. These are not pair-matched the way the rest of this section is – DenseMatcher's appendix table does not state which exact objects or directed pairs it averages over – so this is a category-level sanity check on evaluator-protocol differences, not a controlled comparison.

Category	Pipeline Err (this report)	DenseMatcher Err (reimplemented)	DenseMatcher Err (published [6])
Animals	8.578	14.429	3.21
Chairs	1.658	6.329	3.51
Broccoli	11.313	7.978	3.06
Shampoo	1.043	2.497	3.15

Table 7. DenseMatcher's own published per-category Err [6] against this report's reimplemented graph-geodesic score for DenseMatcher, for the four categories DenseMatcher's paper reports individually. Lower is better. The published numbers are consistently lower (better) than the reimplemented ones on three of four categories, and higher on shampoo – evidence that evaluator and protocol differences, not just method differences, separate this report's DenseMatcher numbers from DenseMatcher's own published ones.

The gap between DenseMatcher's published and reimplemented numbers is not uniform in direction, which is itself informative: it is evidence against a simple "the graph-geodesic evaluator is unfair to DenseMatcher" explanation, and consistent instead with the evaluator and protocol differences already noted above (Dijkstra graph-geodesic distance rather than exact geodesic distance, a different normalized-threshold convention that DenseMatcher's own paper does not fully specify, and – for categories outside the clean-preparation subset – a different mesh orientation).

7.6 Mean vs. median diagnostic

Under the same graph-geodesic scorer, DenseMatcher's released predictions have a much heavier failure tail than their median alone suggests, while the pipeline's tested granularities form a less skewed distribution.

	Pipeline AUC	Pipeline Err	DenseMatcher AUC	DenseMatcher Err
Pooled diagnostic				
Mean	0.8255	4.659	0.6632	12.089
Median	0.8627	3.444	0.7464	6.410

Table 8. Mean vs. median AUC/Err. Pipeline values pool 2328 (directed pair, PF-setting) rows; DenseMatcher values pool 496 unique directed pairs, so this is a tail diagnostic rather than the headline fairness claim.

8. Surface Quality

Dense correspondence metrics ask whether semantic regions match. They do not by themselves say whether the final mesh stays close to the target surface and keeps smooth, locally consistent geometry. For that reason the project also evaluates surface quality using symmetric Chamfer, F@0.02, robust $p95$ surface distance, normal consistency, and semantic-region versions of Chamfer and F-score, computed in a unit-box coordinate system so numbers are comparable across objects of different original scale. These are proximity and smoothness proxies, not a direct count of mesh-validity failures such as self-intersections or flipped triangles: this report does not measure those numerically, and instead controls for them during optimization through the regularizers in Section 5.6 and inspects for them qualitatively through the figures and video showcases referenced throughout (Section 10 discusses the cases where they appear). In this normalization, a threshold of 0.02 is strict: it asks for surface samples to be very close relative to the whole object.

This section reports the *extended* surface-quality evaluation: in addition to the original 4 SAM3D animal pairs and 2 cross-dataset chain pairs, 4 more SAM3D animal pairs (Cat→Chihuahua, Cat→Fox, Goldie→Fox, Goldie→Pug). No DenseCorr3D ground truth exists for any of these self-created meshes, so only geometric surface metrics are reported, not AUC/Err.

Experiment family	n	Chamfer ↓	F@0.02 ↑	$p95$ ↓	Normal ↑	Sem. F@0.02 ↑
SAM3D animal pairs	8	0.0223	0.9354	0.0212	0.9411	0.9142
Cross-dataset chain examples	2	0.0210	0.9304	0.0236	0.8805	0.9104
SAM3D vehicle pairs	5	0.0343	0.7033	0.0334	0.8585	0.6740
DenseCorr3D fair all-animal best-PF stress test	90	0.0568	0.4724	0.1000	0.8284	0.4525

Table 9. Surface quality in unit-box coordinates, extended evaluation. Vehicles are markedly weaker than animals under the identical `pfonly` recipe.

The animal pairs have mean Chamfer 0.0223 and mean F@0.02 0.935: the final deformed surfaces closely cover the target surfaces, and semantic F@0.02 (0.914) tracks the global score closely, meaning the good fit is not only produced by one large body region covering the target. The vehicle pairs, run with the exact same PartField-only recipe, are clearly weaker: mean F@0.02 drops to 0.703 and semantic F@0.02 to 0.674.

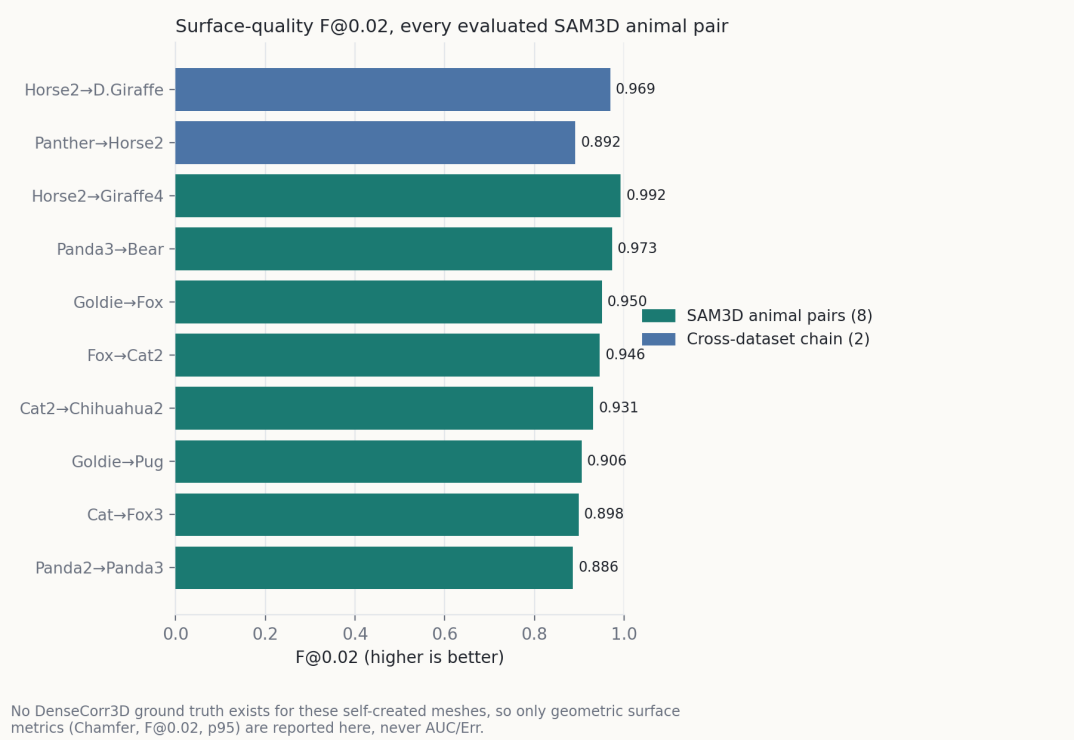


Figure 9. F@0.02 for SAM3D animal and cross-dataset chain pairs.

Run	Chamfer ↓	F@0.02 ↑	$p95$ ↓	Sem. Chamfer ↓	Sem. F@0.02 ↑
Horse2 → Giraffe4	0.0134	0.9923	0.0129	0.0141	0.9836
Panda3 → Bear	0.0199	0.9730	0.0183	0.0210	0.9556
Goldie → Fox	0.0228	0.9503	0.0203	0.0239	0.9306
Fox → Cat2	0.0224	0.9460	0.0209	0.0232	0.9333
Cat2 → Chihuahua2	0.0246	0.9312	0.0242	0.0281	0.9058
Goldie → Pug	0.0248	0.9056	0.0236	0.0283	0.8732
Cat → Fox3	0.0243	0.8985	0.0244	0.0259	0.8818
Panda2 → Panda3	0.0261	0.8859	0.0251	0.0288	0.8495
Horse2 → Dense Giraffe (chain)	0.0189	0.9692	0.0189	0.0209	0.9440
Panther → Horse2 (chain)	0.0231	0.8916	0.0283	0.0243	0.8769
Pink Smart → Red Smart	0.0290	0.8196	0.0269	0.0310	0.7869
F1 car → Verstappen	0.0336	0.7275	0.0364	0.0354	0.7024
Mini Cooper → Pink Smart	0.0348	0.6709	0.0329	0.0370	0.6420
Red truck1 → Red truck3	0.0379	0.6719	0.0371	0.0405	0.6441
G-class → Green SUV	0.0364	0.6268	0.0336	0.0390	0.5946

Table 10. Detailed surface-quality results, sorted within each group by F@0.02.

The best case overall remains Horse2 → Giraffe4, with Chamfer 0.0134 and F@0.02 0.9923: two compatible quadruped structures where PartField separates the main body regions cleanly. The hardest animal case is Panda2 → Panda3, a same-species pair where the lower F@0.02 and higher semantic Chamfer reflect round body regions and pose differences reducing precision.

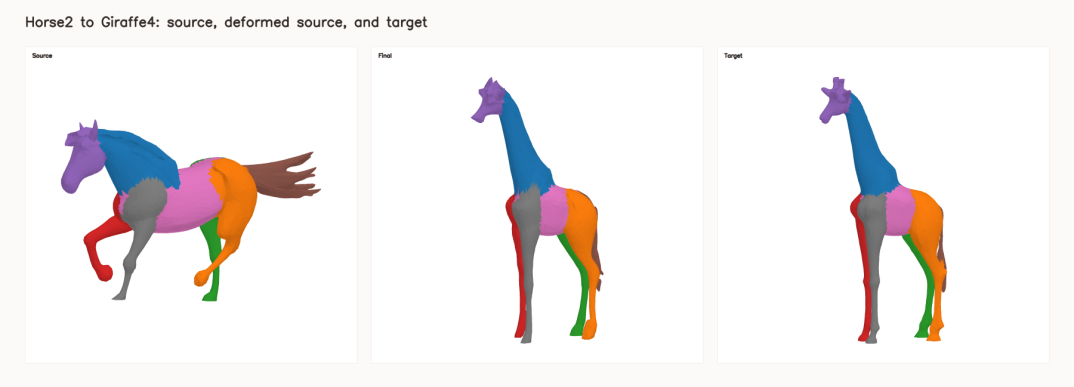


Figure 10. Horse2 → Giraffe4: a compatible large-proportion deformation.



Figure 11. Giraffe → Elephant: hard topology transfer with missing trunk structure.

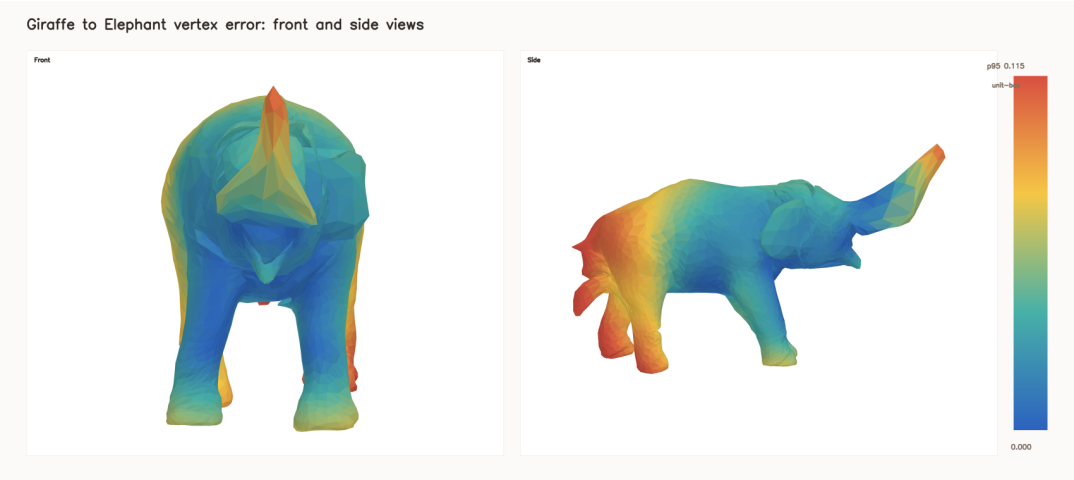


Figure 12. Failure heatmap for Giraffe → Elephant. Color is nearest-target vertex distance in unit-box coordinates, clipped at the run's p95 value shown on the colorbar.

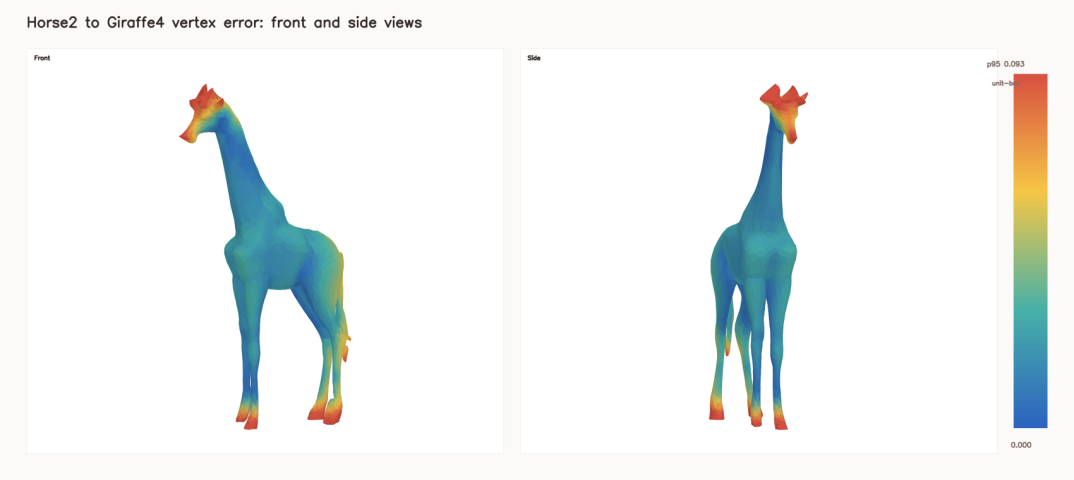


Figure 13. Heatmap for the successful Horse2 → Giraffe4 case. Color is nearest-target vertex distance in unit-box coordinates, clipped at the run's p95 value shown on the colorbar.

9. Cross-Category DenseCorr3D Experiments

This section presents a root-cause diagnosis for the two categories that still lose against DenseMatcher even in the fair, same-evaluator comparison of Section 7: eggplant and broccoli. The same pattern is visible at a glance in Figure 6: categories with reliable, repeated parts perform best (knife, shampoo, cars, chairs), while categories with weak part structure or strong intra-class variation are much harder (broccoli, potato, eggplant).

9.1 Root-cause diagnosis: why eggplant and broccoli lose

Eggplant and broccoli are the two categories that still lose against DenseMatcher in the fair comparison (Section 7), and each has a distinct, identifiable cause.

Eggplant's `groups.txt` divides the mesh into roughly 29 thin rings running along its long axis rather than coarse parts (group-axis correlation -0.996 with the mesh's principal axis, mean ring extent only about 0.10 of the object's length), and PartField's own unsupervised clusters do not track these rings either, so the local Chamfer term has no reliable ring-level anchor to lock onto.

Broccoli's failure has a different cause. Its `groups.txt` follows the crown's irregular, branching floret structure, which PartField's unsupervised clustering does not recover: the buckets it finds do not line up with the florets DenseCorr3D's annotators used, so the local Chamfer term has no reliable structure to guide it.

Appendix C includes additional produce and hammer examples. They are useful qualitative checks, but they are secondary to the main root-cause diagnosis here.

10. Failure Modes

In this project, most bad results fall into a small number of categories: topology mismatch, unstable buckets, orientation or pose mismatch, and metric sensitivity.

Failure mode	Symptom	Typical response
Topology mismatch	Target contains structure the source cannot represent.	Use topology-aware methods or choose more compatible pairs.
Bucket instability	Local loss pulls a source part toward the wrong target region.	Try different PF levels, bucket balancing, or learned bucket matching.
Orientation mismatch	Buckets are swapped or mirrored before optimization starts.	Add canonical orientation and symmetry handling.
Pose mismatch	Limbs or protrusions begin in incompatible positions.	Use staged alignment or pose-aware preprocessing.
Metric tail failure	Mean score looks acceptable but $p95$ is large.	Inspect heatmaps and per-bucket errors rather than only averages.

Table 11. Summary of the main failure modes observed in the project.

10.1 Topology mismatch

The deformation keeps the source topology. This is often desirable because it preserves a clean mapping from original source vertices to deformed vertices. It is also a limitation. A horse can be stretched toward a giraffe, but it cannot truly grow a new topological structure. An elephant trunk, antlers, separated ears, or missing limbs must be approximated using existing source surface area, producing the characteristic failure where the target is globally covered but one semantic part is stolen from a neighboring region.

A smaller but very visible example appears in dog pairs such as Goldie → Pug. The tail tips may be in different positions, and the tip of the source tail does not reliably follow the tip of the target tail: a point-to-point correspondence method could explicitly say that one tail endpoint should match the other, while a PartField bucket mainly says that a tail or rear-body region should match another region, so exact endpoint identity can be lost. Figure 14 makes this concrete by tracking actual vertices rather than buckets. Every vertex is colored by its own rest-pose position (the project's own position-color scheme, `semantic_tracking/vertex_color_tracking.py`: RGB from per-axis normalized XYZ), so the whole body, including the tail's own shape, is legible at a glance rather than reduced to a single highlighted patch on a flat body. On top of that gradient, the farthest-out 30% of the source tail bucket is marked in bright yellow on the undeformed Goldie; because the deformation preserves vertex correspondence exactly, the deformed mesh inherits the identical per-vertex colors, so the same yellow patch shows precisely where that geometry physically ends up – not on the pug's own tail, but pulled onto the flank, near the hip. The pug's own tail tip (found the same way, from its own bucket labels, and colored by its own independent position gradient) is marked yellow on the target as well, so the comparison is the same marker color.



Figure 14. Goldie → Pug tail-tip tracking. The source tail-tip vertices land on the flank/hip rather than the target tail tip.

10.2 PartField bucket instability

PartField buckets are unsupervised. They are useful precisely because they avoid manual labels, but they are not guaranteed to be consistent across large shape gaps. A bucket that represents a leg on one mesh can include body or neck points on another mesh. When this happens, the local Chamfer term can guide the optimizer in the wrong direction.



Figure 15. Poor bucket correspondence in a difficult giraffe-to-elephant pair.

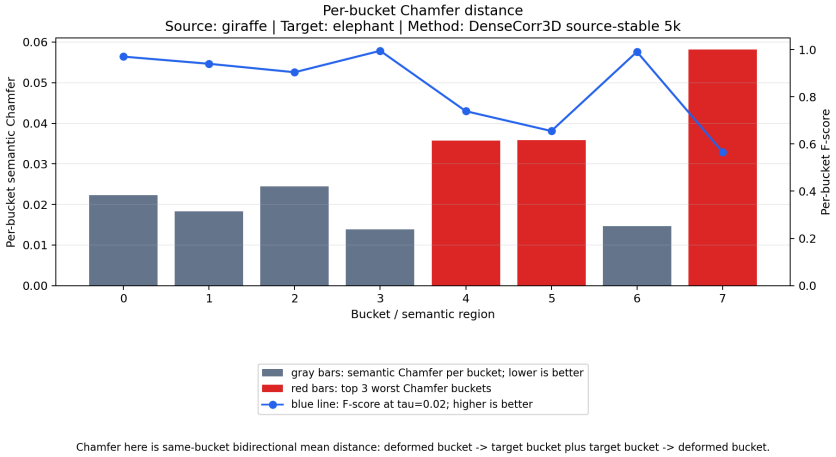


Figure 16. Bucket-level Chamfer reveals which local regions dominate the error.

10.3 Orientation and pose

Many failures are preprocessing failures, not deep failures of the deformation objective. If the source faces forward and the target is rotated, or a category such as zucchini or egg has inconsistent stem orientation, nearest-neighbor and PartField matching can be wrong from the start.

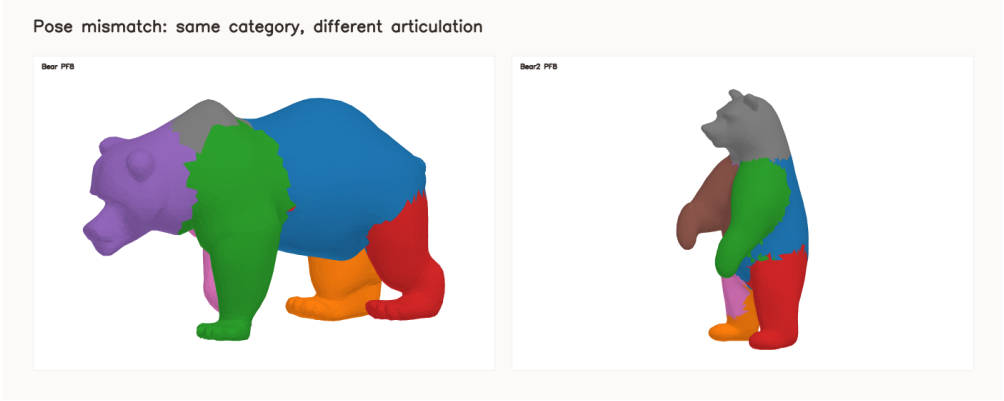


Figure 17. Pose mismatch can make local buckets disagree.

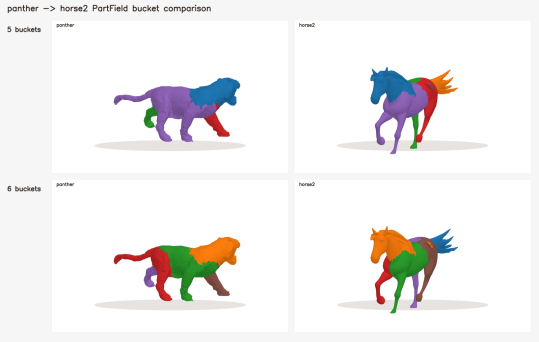


Figure 18. Unaligned orientation can make PartField buckets appear inconsistent.

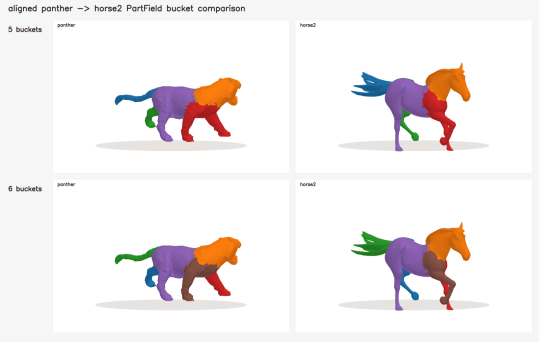


Figure 19. After alignment, the same category often produces more interpretable buckets.

10.4 Metric sensitivity

Metrics measure different properties. Chamfer may be low even when a small part is semantically wrong. AUC can be strong while Err remains high if most points are good but some regions fail badly. F@0.02 at 0.02 is strict and visually meaningful in unit-box coordinates, but sensitive to small sampling and normalization choices. The most useful diagnostic pattern is disagreement between the mean and the tail: if Chamfer is low but $p95$ is high, the result probably has a localized severe failure; if AUC is high but Err is high, many points are correct but some are far away along the target surface; if global F@0.02 is strong but semantic F@0.02 is weaker, the mesh fits the target surface but the local regions are not matching cleanly.

11. Discussion

The strongest result of the project is that local part supervision changes the behavior of mesh deformation in the intended direction, and that this holds up in the graph-geodesic comparison against DenseMatcher's own predictions: on identical directed pairs, with the same scorer applied to both methods, the pipeline wins 21 of 23 DenseCorr3D categories under the range-averaged, zero-best-PF protocol. The optimizer no longer sees the target as only an unordered surface. It sees a weak structural map between source and target. A stricter clean-preparation subset gives the more conservative cross-check: on 12 categories without assisted preparation, it still beats DenseMatcher on 10 of 12 with zero best-PF selection. This supports the same interpretation from the other side: the method benefits from a shared canonical orientation, as the PartField examples in Figures 18–19 make visible, but the main DenseMatcher improvement is not carried only by categories that received extra pose help.

The second important result is that the method is diagnostically useful, in both directions. When it fails, the failure traces to a concrete cause: orientation, bucket instability, topology mismatch, or a category with no real part structure (Section 9.1).

At the same time, the method is not a complete solution to dense 3D correspondence, and the assisted-preparation subset remains a caveat that should be separated from the clean subset. Its deeper, structural limitations – what bounds it, and what would need to change to raise that bound – are addressed directly in Section 12. The right conclusion: PartField local guidance is a strong, audited practical prior for mesh deformation, especially when the source and target share visible, non-rigid part structure, and its limits are as well characterized as its strengths.

12. Limitations

The deformation is only ever as good as the segmentation that guides it. PartField buckets are an unsupervised proxy for correspondence, not correspondence itself: the local Chamfer term only ever knows that "source bucket k should move toward target bucket k," and if that assignment is wrong or unstable – as

Section 9.1 and Section 10 document in detail – no amount of optimization, regularization, or curriculum tuning can recover the correct correspondence from it. This is the ceiling on the whole approach, and it is a structural one, not a tuning problem.

The clearest evidence for that ceiling is what happens when the bottleneck is removed. As a diagnostic, this pipeline was also run on TOSCA [12] pairs spanning five categories (cat, dog, gorilla, horse, wolf) using each pair's real, dense, ground-truth vertex correspondence as the only guidance signal, in place of PartField buckets entirely (the same Jacobian deformation machinery, with the local-bucket Chamfer term replaced by a direct per-vertex correspondence loss). Figure 20 shows three of these pairs – dog, gorilla, and horse – each with a large pose change: in every case the deformed source is visually almost indistinguishable from the target. Scored with the same graph-geodesic evaluator used throughout this report, and averaged over every one of the 23 completed TOSCA dense-correspondence runs in the project (not a selected subset), these runs reach mean AUC 0.950 and mean Err 1.40 (best pair: wolf0→wolf2, AUC 1.000/Err 0.001; weakest: cat0→cat2, AUC 0.786/Err 5.72), clearly above this pipeline's own headline PartField-guided numbers (AUC 0.824/Err 4.77 across all 23 DenseCorr3D categories, Table 5). This is not a claim that dense correspondence is free – TOSCA pairs come with correspondence already solved, which is exactly the hard problem PartField is standing in for – but it suggests that an important remaining gap comes from the quality of the correspondence signal feeding the deformation, not only from the deformation representation itself.

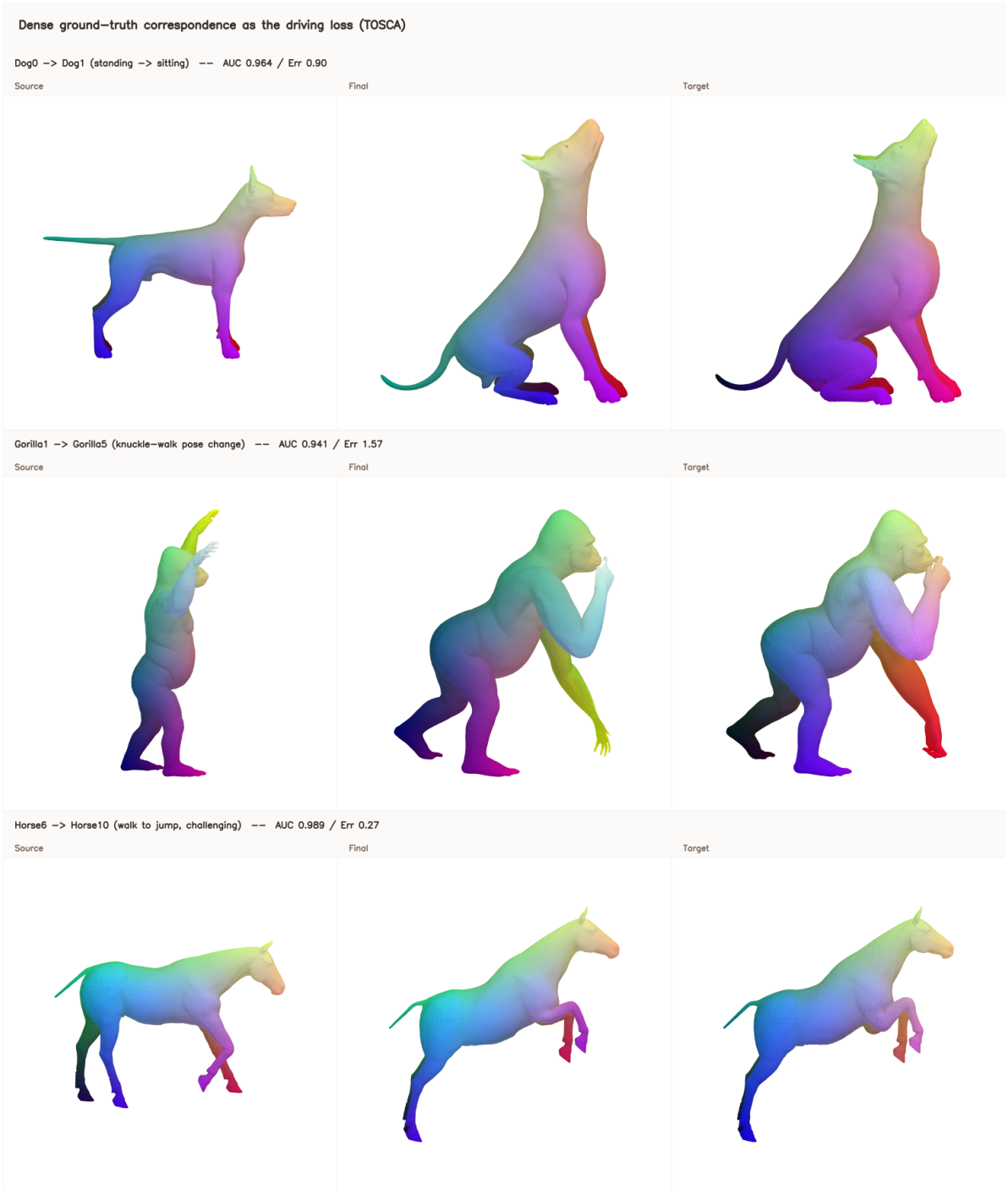


Figure 20. TOSCA dense-correspondence ceiling test: with ground-truth guidance, the deformation representation is usually not the bottleneck.

Category	Completed TOSCA dense-correspondence-guided runs
cat	cat0→cat1, cat0→cat2, cat1→cat4, cat2→cat5, cat3→cat9
dog	dog0→dog1, dog0→dog2, dog1→dog3, dog2→dog5, dog6→dog8
horse	horse0→horse5, horse0→horse6, horse5→horse7, horse6→horse10, horse14→horse15
wolf	wolf0→wolf1, wolf0→wolf2, wolf1→wolf2
gorilla	gorilla1→gorilla5, gorilla1→gorilla8, gorilla1→gorilla14, gorilla5→gorilla8, gorilla8→gorilla14

Table 12. All 23 completed TOSCA dense-correspondence-guided runs used for the aggregate ceiling diagnostic. Figure 20 renders three representative examples from this list.

Fixed topology: no new parts can be created. The deformation moves an existing mesh; it cannot introduce a new limb, horn, or disconnected component that the source does not already have some surface area to repurpose. When the target genuinely needs one – an elephant's trunk, a giraffe's ossicones, a missing limb – the pipeline can only approximate it by stealing area from a neighboring source part (Section 10), which is part of why the SAM3D vehicle results in particular do not close the gap: rigid, sharply-creased geometry has essentially no slack part to repurpose this way.

Vertices are moved, never deleted. The Jacobian-field parameterization (Section 5.5) changes vertex positions; it has no mechanism to remove a vertex or a face. If the target is smaller or simpler than the source in some region, that source geometry cannot disappear – it can only be compressed, flattened, or folded into a thinner sliver, which is a recurring source of the self-intersection and surface-quality problems documented in Section 8.

13. Future Work

Several extensions follow naturally from the experiments.

1. **Closing the segmentation ceiling with real dense correspondence.** Section 12 suggests that replacing PartField buckets with real dense correspondence substantially raises the achievable ceiling (23 TOSCA pairs: mean AUC 0.950 vs. this pipeline's own 0.824 headline). Learning or acquiring a real dense correspondence predictor, rather than relying on unsupervised segmentation matching, is the single change most likely to improve results beyond what better buckets or more regularization can achieve.
2. **Learned bucket matching.** A smaller step in the same direction: instead of assuming bucket k matches bucket k, learn a soft transport matrix between source and target buckets.
3. **Learned bucket-count selection.** The LLM-based bucket selector (Section 5.1) was built for pipeline completeness rather than as a benchmarked component; replacing its prompt-based reasoning with a selector trained on which K actually performs best for a given pair is a natural next step.
4. **Topology-aware deformation for rigid domains.** The vehicle results in Section 8 suggest that fixed-topology deformation is a worse fit for rigid, sharply-creased bodies than for articulated ones; remeshing, implicit surfaces, or panel-aware segmentation are more promising directions there than further loss-weight tuning.
5. **Category-level evaluation with a fixed protocol.** Convert the strongest selected-pair results into a benchmark protocol with predetermined PF levels and no per-pair model selection, matching the zero-selection standard already applied to the DenseMatcher comparison.
6. **Better semantic diagnostics.** Report bucket transport entropy, per-part failure curves, and confidence maps so bad correspondences can be identified automatically rather than by manual inspection.

14. Conclusion

This project built and evaluated a semantic-guided part-aware mesh deformation pipeline, after redirecting away from an earlier, heavier image-semantic approach built on MeshUp (Appendix B). Starting from a

source and target mesh, the method uses PartField buckets to turn global shape matching into a more structured problem: the whole surface should match, but corresponding local parts should match as well. Evaluated against DenseMatcher's own predicted correspondences on identical directed pairs, using the same graph-geodesic evaluator applied to both methods, the method wins 21 of 23 categories with predeclared category-specific PF ranges and no best-PF picking within those ranges. It gives clean surface geometry on articulated self-created SAM3D pairs, exposes weaknesses and traces its failure cases down to specific, identifiable causes – orientation, unstable buckets, topology mismatch, or evaluator sensitivity. The main contribution is therefore a practical, interpretable, and audited deformation pipeline: simple enough to analyze, structured enough to avoid many of the semantic mistakes made by pure Chamfer optimization, and evaluated with the same scrutiny applied to its strongest results as to its weaknesses.

A. Video Showcase

An interactive video showcase accompanies this report: hover-to-play deformation clips for a range of category pairs, alongside the headline comparison numbers against DenseMatcher from Section 7. It is available at `https://rona-latifaj-uni-freiburg.github.io/semantic-mesh-deformation/`.

B. Earlier Work: Adding Semantic Awareness to MeshUp

Before the project became a direct mesh-to-mesh deformation pipeline, the initial direction was to make MeshUp more semantically aware. MeshUp is a multi-target mesh deformation method that uses score distillation and differentiable rendering to guide a source mesh toward image-level target concepts [1]. This is powerful because it can use image priors, but it is not inherently a dense 3D correspondence method. The optimizer sees rendered images and global shape regularization. It does not automatically know that the tip of one tail should follow the tip of another tail, or that a leg should correspond to a leg across different animals.

The motivation for the first set of experiments was therefore simple: add semantic feature signals to the MeshUp loop. The project explored DINO-style visual features, DINO PCA visualizations, semantic vertex correspondences, and a preliminary RAE-DiT teacher idea. In the image-feature branch, the mesh was rendered from several views, the rendered images were passed through DINOv2-style feature encoders [3], and the resulting feature maps were used as semantic guidance. In the diffusion branch, the original MeshUp-style score-distillation setup was still present, using the same broad idea as DreamFusion: optimize a 3D object through gradients obtained from a 2D generative prior [2]. The tracked configuration used an XL DeepFloyd IF stage, a large cascaded text-to-image diffusion model; the public model card lists the Stage-I XL model at about 4.3 billion parameters [4].

This direction was scientifically reasonable, but it had a practical problem: the semantic signal remained indirect. DINO features are extracted from rendered views, not from stable 3D surface points. A feature that appears reliable in one camera can become ambiguous under occlusion, pose change, lighting, or a different silhouette. Similarly, diffusion gradients can encourage a plausible rendered image without producing a clean point-to-point or part-to-part surface correspondence. In other words, the old pipeline was trying to obtain 3D semantic deformation from 2D semantic supervision.

Run	Epochs	Main guidance	Time	Peak RAM	FID ↓	Interpretation
A1 baseline dev	2500	SDS, no DINO	24:10	12.5 GiB	605.19	Reference dev run.
A1 DINO dev	2500	SDS + DINO	24:49	12.1 GiB	592.20	DINO improved FID by 12.99 points, about 2.15%, but the gain was small.
A1 baseline non-dev	2500	SDS, no DINO	28:00	14.9 GiB	494.86	Stronger baseline run without DINO.
A1 DINO non-dev	6000	SDS + DINO	56:13	16.4 GiB	503.35	Longer DINO run did not beat the shorter non-DINO baseline.

Table 13. Early MeshUp semantic-guidance ablation. FID was useful as an image-level diagnostic, but not a direct 3D correspondence metric. The DINO gain was small in the controlled dev comparison and disappeared in the longer non-dev comparison.

The compute profile reinforced this conclusion. The tracked MeshUp configuration used `batch_size=8`, `accum_iter=2`, and an XL diffusion model. At 2500 epochs this corresponds to roughly 80,000 diffusion U-Net batch-item evaluations; at 6000 epochs, roughly 192,000. The DINO side was also not free: with eight rendered views at approximately 512 pixels, the feature extractor handled about 10,368 patch tokens per mesh state before any matching or loss computation. The old jobs were submitted with large memory envelopes, including 40 GB for the ablations and a 64 GB DINO presentation script – compare Section 5.8, where the final PartField pipeline's representative runs measured under 1.8 GiB peak RAM.

Branch	Main signal	Evidence from the project	Outcome
SDS + DINO MeshUp	Rendered-image diffusion gradients plus rendered DINO features.	Small dev FID gain; longer DINO run worse than shorter baseline; 40–64 GB job envelopes; tens of thousands of diffusion evaluations.	Useful exploration, but too expensive and too indirect for the main report.
Semantic vertex correspondences	PartField descriptors, bounding-box position, normals, and optional labels used to map source vertices to target vertices.	Dachshund → cat produced 3111/4712 unique target vertices and 0.9291 PartField label agreement, but many vertices still collapsed to repeated targets.	Good diagnostic signal, but direct semantic losses damaged geometry in difficult cases.
Semantic reindexing after topomatch	Use clean geometry from topomatch, then reorder vertices according to the semantic map.	Dog-pair label agreement ranged from about 0.70 to 0.86, with exact geometric recovery when the operation was pure reindexing.	Useful for correspondence analysis, but not a new deformation force and only exact for matching topology.
Current PartField mesh-to-mesh deformation	Global Chamfer, local bucket Chamfer, bucket balancing, and regularization directly in 3D.	Representative final jobs ran in about 4.6–17.5 minutes, with measured peak RAM below 1.8 GiB.	Chosen direction: simpler, more interpretable, and evaluated with direct 3D metrics, now including a same-evaluator comparison against a real correspondence method.

Table 14. Why the project pivoted from image-semantic MeshUp guidance to direct mesh-to-mesh deformation. The new pipeline does not give up on semantics; it moves semantic structure into explicit 3D part supervision.

The final method still tries to deform semantically meaningful parts, but it does so through PartField buckets, global and local Chamfer losses, bucket balancing, regularization, and 3D evaluation. This made the experiments easier to interpret, and, ultimately, made it possible to run the much stronger same-evaluator comparison against DenseMatcher in Section 7 – a comparison that would not have been meaningful for a method whose main signal was 2D-rendered image similarity.

C. Additional Qualitative Results

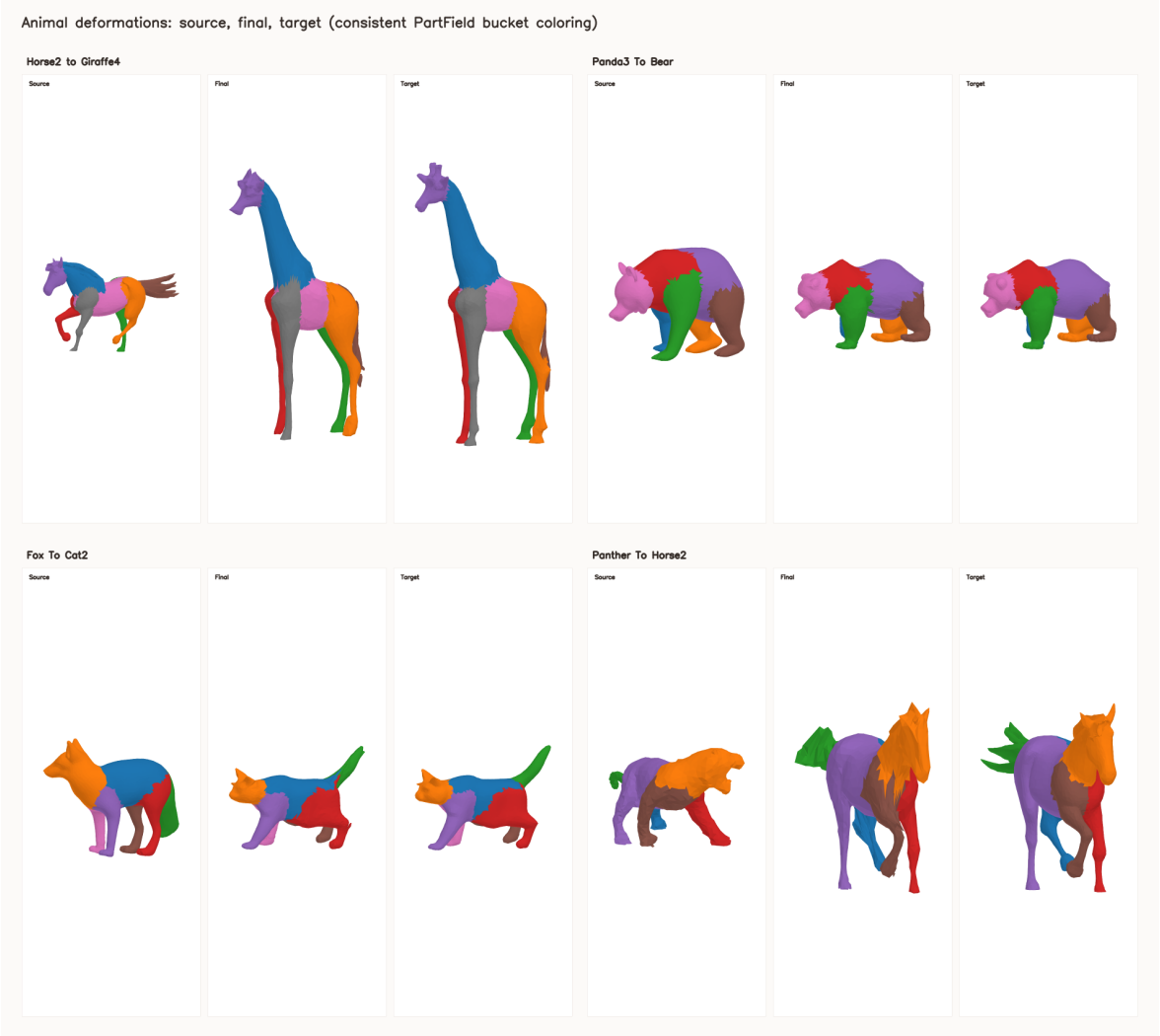


Figure 21. Animal deformation showcase with consistent discrete PartField bucket coloring.

Figure 21 summarizes why animals are both compelling and difficult. The method can stretch a horse toward a giraffe, a panda toward a bear, a fox toward a cat, or a panther toward a horse while keeping local regions coherent. At the same time, animals contain limbs, tails, necks, and heads whose correspondence depends on pose, which is why animal results need both numerical evaluation and visual inspection.



Figure 22. Hammer deformation: a stable handle/head structure gives PartField clear anchors.

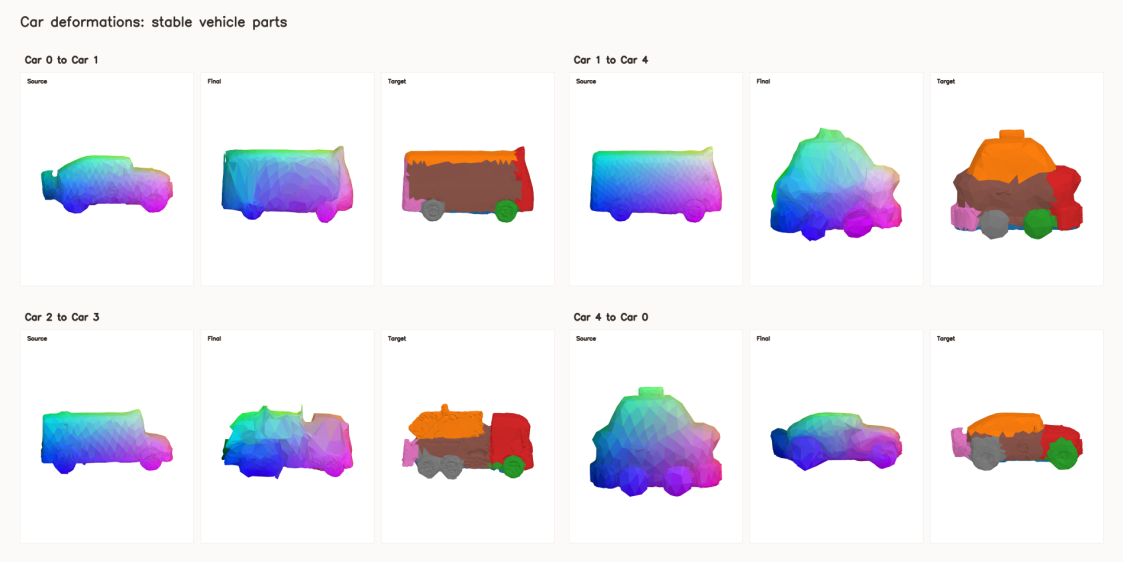


Figure 23. Car deformation showcase on the DenseCorr3D benchmark category. Source and final panels use inherited source-position coloring; target panels use PartField bucket coloring, so colors are not direct cross-panel correspondence labels in this figure.

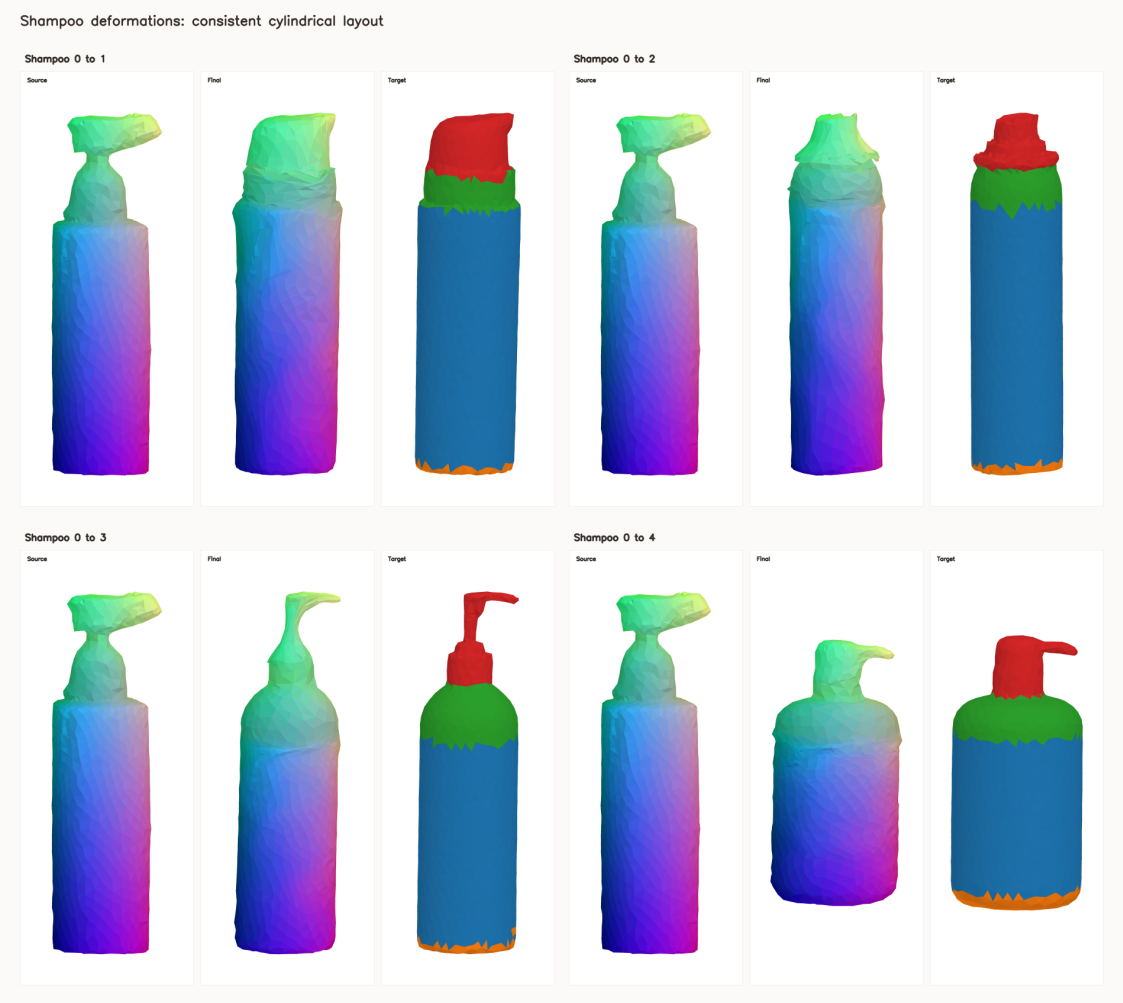


Figure 24. Shampoo deformation showcase. The tall body/top structure is a clean PartField case. Source and final panels use inherited source-position coloring; target panels use PartField bucket coloring, so colors should be read as within-panel structure rather than direct cross-panel labels.

D. Local Chamfer Spike Failure Example

Section 5.6 explains why an evenly-weighted, per-point local Chamfer term can pull an unclaimed source point toward its own single nearest target point at full strength, with nothing else to keep it near its neighbors. Figure 25 shows a concrete instance of this from an early exploration of the bulldog→dachshund pair, using a PartField chamfer-only objective ($\lambda_g = 0$, no jump/edge-stretch regularization, and neither the asymmetric source-to-target weighting nor the unmatched-point down-weight from Section 5.6 enabled yet). The front leg collapses into a thin spike reaching well past the target's own leg length, and the head develops a smaller spike above the ear. Introducing $\alpha < 1$ and the unmatched-point down-weight γ (Section 5.6) reduced this failure mode in later runs.

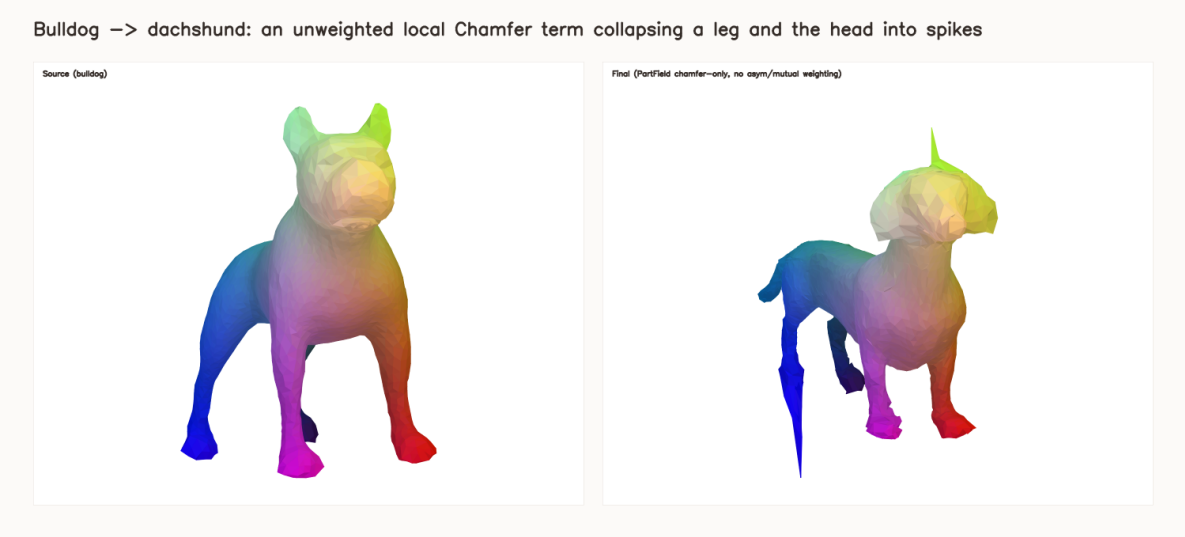


Figure 25. Bulldog→dachshund, PartField chamfer-only, before the asymmetric and unmatched-point weighting described in Section 5.6 was introduced. The front leg and the head each collapse into a thin spike rather than deforming smoothly. Coloring is inherited source-position, consistent with the other deformation figures in this report.

E. PartField Granularity by Category

Section 7 deliberately reports headline numbers under zero PF selection (Table 6's "PF range" column) rather than a single best-per-category granularity, precisely to avoid quietly picking a winner. That said, the question of which single PartField granularity performs best per category is useful diagnostic information in its own right. PF01 is the coarsest setting: it gives one bucket, so it provides almost no part separation. Higher PF values create more buckets and can express more detailed part structure, but the extra detail only helps when the shape really contains stable parts. Table 15 reports the oracle best-single-PF value for every category, computed the same way as the "best PF per category" rows in Table 5 (AUC/Err at that pair's single best-scoring granularity, not averaged across the range).

Table 15. Best-scoring single PartField granularity per DenseCorr3D category (oracle, not the zero-selection headline).

Category	Best PF	AUC	Err
animals	PF06	0.7443	7.432
apple	PF05	0.8796	3.013
backpack	PF04	0.8895	2.834
banana	PF01	0.8607	3.485
bell_pepper	PF03	0.8056	5.079
broccoli	PF02	0.6361	9.472
carrot	PF04	0.7173	7.273
cars	PF06	0.9476	1.332
celery	PF06	0.9046	2.386
chairs	PF08	0.9426	1.528
cucumber	PF01	0.8287	4.299
egg	PF03	0.9088	2.279
eggplant	PF01	0.8159	4.601
knife	PF02	0.9731	0.672
mushroom	PF02	0.7986	5.322
onion	PF02	0.8384	4.054
peach	PF01	0.8984	2.540
pear	PF02	0.8370	4.076
potato	PF01	0.7829	5.428

Category	Best PF	AUC	Err
pumpkin	PF02	0.8785	3.037
shampoo	PF03	0.9701	0.748
tomato	PF02	0.9110	2.224
zucchini	PF01	0.8797	3.008

The pattern in Table 15 is fairly clean once the direction of PF is kept straight. Categories with distinct, repeated, or articulated parts often prefer intermediate to finer settings – chairs (PF08: seat, back, legs, stretchers), cars (PF06), celery and animals (PF06), backpack and carrot (PF04) – because one or two buckets cannot represent their real structure. Smooth or weakly differentiated produce often prefers very coarse settings, usually PF01–PF02 (banana, cucumber, eggplant, mushroom, onion, peach, potato, pumpkin, zucchini), because additional buckets split the surface into arbitrary rings or patches that do not correspond reliably across instances. Broccoli is the exception that explains the limit: its best setting is PF02, but even that cannot recover the irregular floret groups used by DenseCorr3D's annotation.

F. Curriculum Ablation Diagnostic

The optimization is staged because the problem is non-convex. If fine part buckets are imposed too early, the source can get trapped in a bad local minimum. If only coarse/global alignment is used for too long, the optimizer may find a geometrically plausible but semantically wrong solution. Because the ablation covers only 8 hard DenseCorr3D pairs, it is treated as a supporting diagnostic rather than as part of the main benchmark claim.

Stage	Purpose	Typical signal
Coarse alignment	Move the source into the target's overall pose and scale.	Coarse surface and PartField guidance.
Part-aware fitting	Align corresponding local regions.	Local bucket Chamfer, bucket balancing, PF6–PF8.
Regularized refinement	Improve surface fit without destroying mesh quality.	Lower learning rate, regularization, edge and Jacobian penalties.

Table 16. The staged deformation schedule used in the curriculum diagnostic.

On 8 hard DenseCorr3D pairs, the three-stage curriculum improved both metrics over a single-stage baseline on the majority of pairs (Table 17): mean AUC rose from 0.804 to 0.823 (6/8 pairs improved) and mean Err fell from 6.05 to 5.60 (6/8 pairs improved).

Pair	Baseline Err	Curriculum Err	Baseline AUC	Curriculum AUC
Elephant → Giraffe	7.486	6.320	0.803	0.856
Elephant → Moose	5.856	5.172	0.799	0.830
Giraffe → Bear	0.778	0.805	0.969	0.968
Moose → Cheetah	7.138	6.740	0.740	0.763
Moose → Giraffe	10.465	9.623	0.727	0.748
Moose → Panther	8.385	7.692	0.694	0.718
Panther → Giraffe	2.150	2.078	0.916	0.926
Panther → Moose	6.144	6.360	0.781	0.778
Mean	**6.050**	**5.599**	**0.804**	**0.823**

Table 17. Three-stage curriculum vs. single-stage baseline, 8 hard DenseCorr3D pairs.

G. Evaluator Design Notes

Because DenseCorr3D's official evaluator and its per-mesh geodesic distance matrices were never published, Section 7 implements its own graph-geodesic (Dijkstra) scorer rather than reproducing DenseMatcher's published benchmark protocol; both methods are then re-scored under that one evaluator, applied identically to both sides. A couple of design choices were tried before settling on the version used throughout this report: scoring each directed pair with its own matched direction (`surface_map` vs. `surface_map_inv`) rather than averaging both directions together, and using DenseMatcher's released correspondence map directly rather than an ICP-refined or adjoint-refined variant of it. The production evaluator uses `surface_map`, matched per pair by direction, exactly as described in Section 7.

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